CHAGEE

CHAGEE SINGAPORE

THE OPPORTUNI-TEA — UNLOCKING GROWTH AT CHAGEE SG WITH AI

OPIM639 · GROUP 3 · KEZIAH, JIN, AYUMI, REVAN · OCTOBER 2026



AGENDA

Spilling the Tea

03 **Introduction**
Chagee Singapore, our scope and why

04 **Executive Summary and Business Plan**
Two key challenges and one strategic decision

05–11 **Our Business Recommendations**
Dynamic demand management · AI-enabled in-store planning · new-store site selection

12 **Execution Plan**
Three months to plan, then a phased six-month launch

13 **Financial Impact**
What the three proposals are worth over five years

14 **Q & A**

\+ **Appendix**
Earlier slides, derivations, tables, sources and the Australia extension



Today, we're presenting Chagee

A tea brand most of you know: order in the app, collect at the counter, or deliver to your location.



Our scope

Chagee Singapore's management and its customers

Outlets are **company-owned, not franchised**, so the counter, roster and equipment are Chagee's own calls.

Why Chagee?

1. **Many stores, managed by one company**, one data system, with lots of data appendix →
2. **App-centric**: collects personal and transactional data
3. **Fast-growing, playful**: rapid expansion, room for improvements with AI
4. **An insider's view**: we were fortunate to interview a staff member

FROM FARM TO CUP · OUR PROJECT FOCUS IS DOWNSTREAM

Source
Yunnan tea; local milk and fruit

Warehouse
Restocking from forecasts

Upstream data and decisions sit with **Chagee corporate**

Outlets
Make and hand over

via Chagee App

Last mile
Collection or delivery

Downstream data is analysed and decisions are made by **Chagee SG** (APAC HQ)

OUR FOCUS | **The downstream operational challenges: some of the most pressing issues for Chagee Singapore's management.**

Our business plan to solve Chagee's problem statements



CASE STUDY · OUR INTERVIEW

"Mei"

Membership Operations · APAC, Chagee

Persona: name changed, interviewee anonymised

WHAT MEI TOLD US

Through the interview, we learnt about Chagee's **business model**, its **local operations**, and some of its most **pressing concerns and strategic needs**, which AI could potentially help solve.

TWO KEY CHALLENGES + ONE STRATEGIC DECISION POINT

KEY CHALLENGE

Online retailing

Use data for better demand management, to grow sales and manage the customer experience.

PROPOSAL 1

Dynamic demand management

At peak, AI reroutes pickups and deliveries to nearby outlets with room, decreasing drop-offs and wait times, and increasing overall sales.

See how it works →



DECISION-MAKER

Zone and branch manager

KEY CHALLENGE

In-store management

A faster, smoother counter: shorter waits, a crew matched to demand and less wastage.

PROPOSAL 2

AI-enabled in-store planning

Higher-precision outlet and zone forecasts help managers decide on staffing, tea machines and fresh stock to maximise store performance.

See how it works →



DECISION-MAKER

Zone and branch manager

STRATEGIC EXPANSION DECISION

New store openings

Where to open next, so a new store is profitable and adds customers instead of moving them.

PROPOSAL 3

New-store site selection model

Picks new sites that add customers and profit.

See how it works →



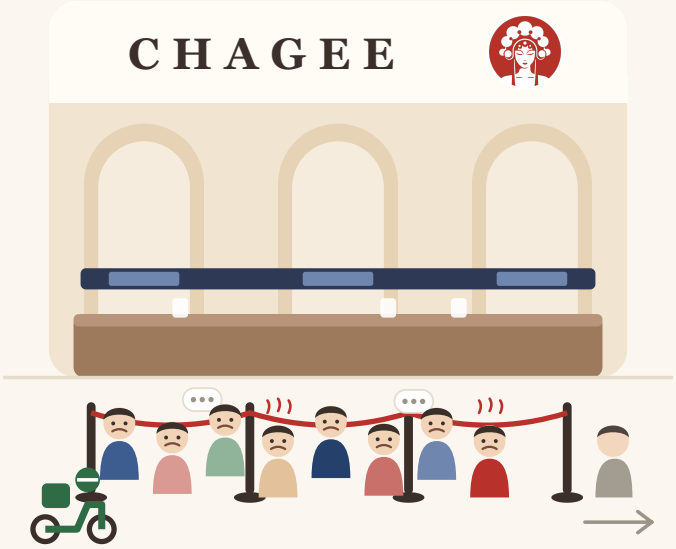
DECISION-MAKER

Expansion and growth manager

Proposal 1: Dynamic demand management

KEY CHALLENGE 1

Demand management: crowded outlets lose sales, with impacted customer experience



WHAT IT COSTS CHAGEE TODAY

Drop-offs
Customers see the wait and leave without ordering.

Crowding
Those who stay crowd the outlet, so it sells less and delivery orders slow down.

Long waits
Customers wait longer than they expected; the experience suffers.

Delivery orders arrive blind
Delivery customers cannot see the in-store queue, so orders keep coming and riders wait at the counter.

Spoils the in-store image
Chagee's elegant stores are part of the brand. A crowd of frustrated customers at the door undoes the classy experience the brick-and-mortar model is built on.

With AI, this can change

- **Higher sales**
  Demand spread across nearby outlets instead of lost.
- **A better brand image**
  Faster turnaround, well managed, a smart system.
- **Calmer counters**
  Delivery riders do not wait long or crowd the counter.

From Chagee's data to demand spread across outlets

An AI agent uses predictive AI to recommend, generative AI explains, and people make the final call.

1 The data

What Chagee collects, or could collect

Chagee app & sales
Orders by outlet and time · Customer location · Margin per drink

Outlet operations
Live queue of open orders · Crew on shift · # of machines

Delivery apps
Delivery orders · Rider waits

Historical events
Decisions made · Profitability outcomes

2 Proposed AI agent

Trigger, recommend, decide, with guardrails

TRIGGER

Wait runs long
A store's wait is, or is about to go, above 40 min.

AI AGENT

Recommend outlet, action, incentive
Predictive AI on live store data and patterns learnt from past decisions, within approved rules.

PEOPLE DECIDE

Customers and managers choose
Generative AI explains each recommendation; they accept or decline.

3 What it does

Example of actions at peak, and it keeps learning

Offers a faster outlet
Routes pickups to the outlet that can hand over sooner.

Recommends a delivery pause
Recommends pausing third-party delivery orders and redirecting demand; the zone manager reviews and approves.

CONTINUOUS LEARNING

Gets smarter over time

Every new decision and its outcome feed back in, so future decisions keep improving.

THE OPERATIONS IDEAS BEHIND IT

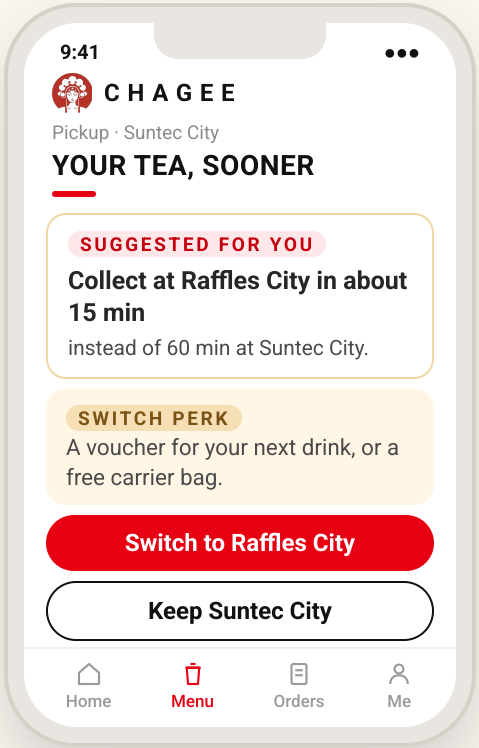
Demand shaping	moving demand strategically
Capacity pooling	with nearby outlets
Reducing bottlenecks	dynamically

KEY BENEFITS

- ✓ **Balances the zone**, thereby increasing sales
- ✓ **Learns from past decisions and live data from stores** to raise profit while enhancing customer experience

Illustration of proposal 1: customers are steered to the outlet with room

PICKUP · CHAGEE APP



A faster outlet, offered

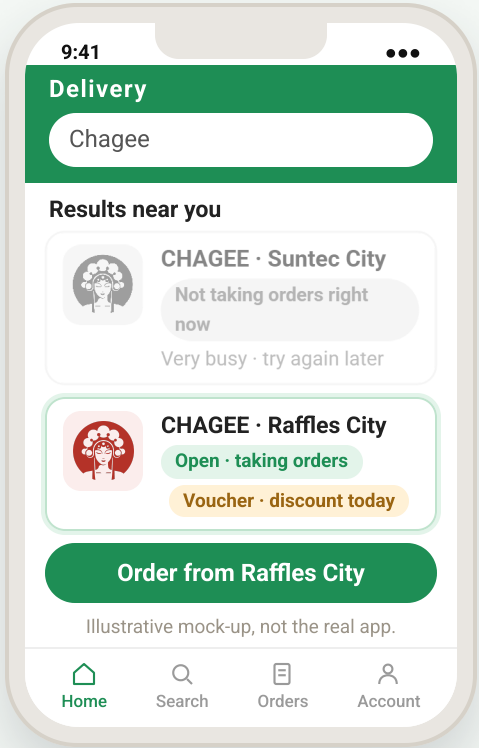
Suntec City is swamped, so the app suggests Raffles City and explains why.

45 min sooner
collect in about 15 min instead of 60

DYNAMIC PRICING
A voucher or a carrier bag nudges and rewards the switch, offered only while Suntec City is swamped.

Customer chooses before paying.

DELIVERY · E.G. GRABFOOD



The busy outlet steps back

The agent recommends pausing Suntec City on delivery apps and redirecting demand; the zone manager reviews and approves.

Orders flow to Raffles City
no riders stuck at the busy counter

DYNAMIC PRICING
Instead of closing Suntec City on delivery altogether, a voucher at Raffles City can draw demand to the outlet with room.

Zone manager approves the pause and redirection.

Execution steps →

Proposal 2: AI-enabled in-store planning to achieve high-performing outlets

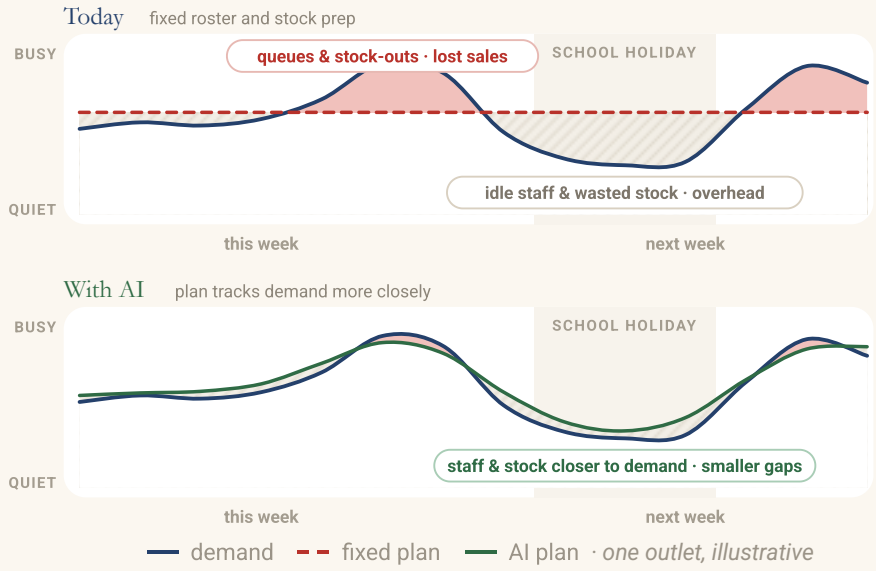
KEY CHALLENGE 2 In-store management: matching staff, machines and stock to daily demand

IN PLACE TODAY *Chagee annual report, FY 2025*

- Group-level demand forecasting for stock and inventory
- Centralised, digital store management

ENVISIONED MODEL *enhanced with predictive analytics and AI*

- Forecasts by outlet per day
- AI-recommended rosters, staff moves and machines to run
- Management of fresh inventory, less waste
- Unoptimal store set-ups flagged for review



With AI, this can change

- **Fewer lost sales** — Ready for busy days.
- **Less idle staff** — Rosters flex on quiet days.
- **Less wasted stock** — Fresh prep matches demand.
- **More optimal machinery count** — How many to match demand.

Managers can use insights to plan and manage stores better

HOW IT WORKS

1. **DATA** — examples: orders · staffing · machines
2. **PREDICTIVE AI** — forecasts each outlet by day
3. **AI PLATFORM** — provides recommendations and a chatbot
4. **MANAGERS** — view insights and plan the week

Chagee Store Insights Dashboard – Zone View — Updated 7am daily

STAFF FOR THE COMING WEEK

OUTLET	RECOMMENDATION
Raffles Place	Reassign 1 crew → City Hall
Suntec City	Maintain current crew
City Hall	Receive 1 crew from Raffles Place

WHY School-holiday pattern learnt from past years: office areas like Raffles Place quieten, leisure areas like City Hall get busier.

UPCOMING PEAKS & QUIET SPELLS

▲ Peak	Suntec City · Fri to Sun, afternoons
▲ Peak	City Hall · weekdays, school holiday
▼ Quiet	Raffles Place · Mon to Fri, school holiday

TEA MACHINES · IN STORE

	CURRENT	RECOMMENDED
Raffles Place	4	3 ▼
Suntec City	4	4 =
City Hall	3	4 ▲

FRESH STOCK · PACKAGES PER WEEK

	CURRENT	RECOMMENDED
Raffles Place	9	7 ▼
Suntec City	8	9 ▲
City Hall	6	8 ▲

One model that keeps getting smarter

1. **Events feed in** — Rerouted pickups, delivery pauses and vouchers from Proposal 1
2. **The model retrains** — on these events and the demand they shifted
3. **Forecasts sharpen** — so this dashboard anticipates better, even after protocol changes

Over time, Chagee manages its stores better, ahead of demand rather than reacting to it.

Execution steps →

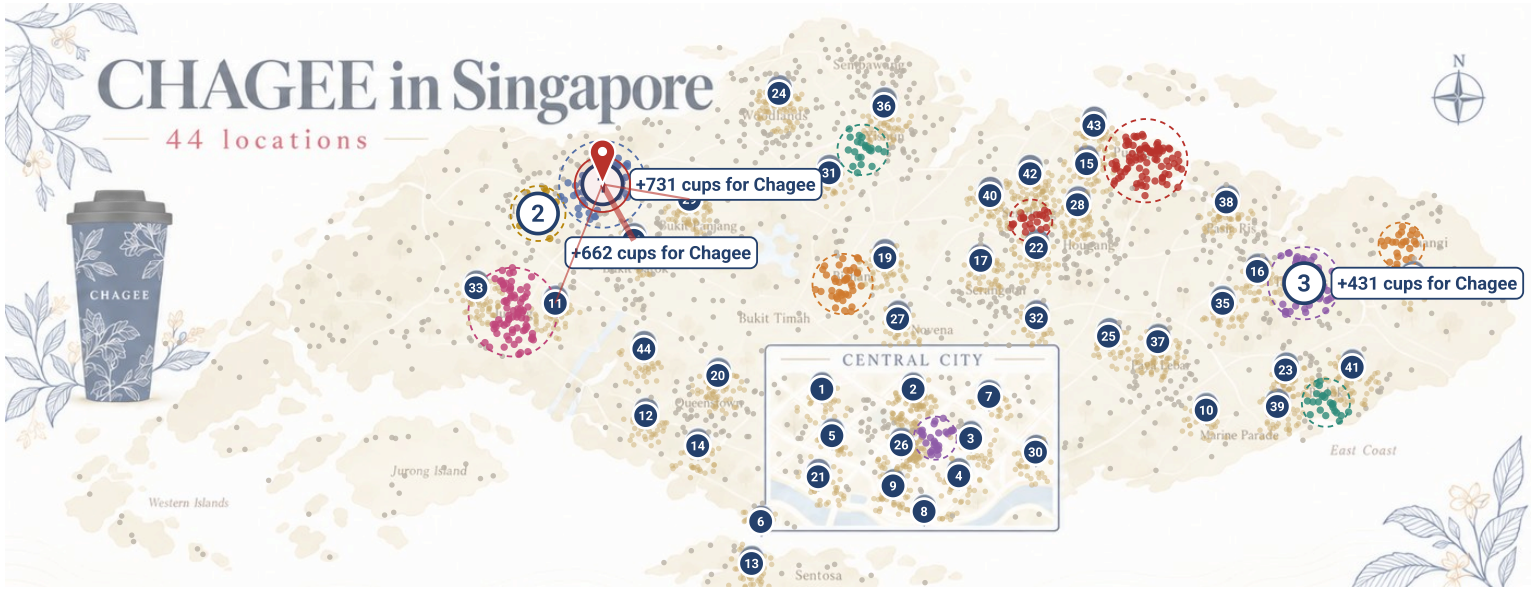
THE OPERATIONS IDEAS BEHIND IT

- **Demand forecasting** — by outlet and day
- **Capacity planning** — staff, machines and stock to the forecast
- **Throughput** — find and fix the slowest step
- **Staff cross-training** — crew cover different roles / outlets

Open where an outlet adds customers, not where it moves them

BUSINESS PROBLEM A new outlet may look successful while mostly moving sales from nearby stores, while other customers have no convenient outlet. Context Chagee aims to expand rapidly in Singapore; the number of stores is confidential (Chagee source, anonymous). Simulated searches below.

NEW-STORE FORECAST



SUGGESTED SITE 1 · 2.7 KM FROM 34 · WEST MALL

Strong site: mostly new cups

+731 cups a day for Chagee

The new store sells 837 a day (1.5× an average outlet). 106 of those move from outlets nearby:

34 · West Mall, 2.7 km	−77
29 · Bukit Panjang Plaza, 3.3 km	−28

Peak hour 13:00: **92 cups** → **2 people** on today's counter (appendix).

Synthetic app data on the team's map; outlet positions approximate (the central inset is enlarged). Not a trained model: the point is the method. Its five-year P&L: appendix.

TAKEAWAY | **Judge a site by the cups it adds to Chagee, not the cups it sells: the cluster near Choa Chu Kang adds about +731 a day; beside an existing outlet, the same store adds +216.**

Investigate the site that adds customers, then check the downside

THE SAME NEW OUTLET AT TWO SITES · SYNTHETIC APP DATA · FIVE YEARS, OUR ASSUMPTIONS

CRITERION	BESIDE 34 · WEST MALL OUR ASSUMED BASELINE	CLUSTER NEAR CHOA CHU KANG DBSCAN + DEMAND MODEL
Customer access: cups from people with no outlet within walking distance	47%	**98%**
Cups a day the store sells	604	837
Net additional cups a day, after cups moved from nearby outlets	+216	**+731**
Incremental operating profit, five years, base volume	−S$0.6m	**S$1.9m**
Investment at opening	S$0.4m	S$0.5m incl. AI share
Payback	not within 5 years	about 1.4 years
Downside: low volume, five years	−S$1.2m	S$0.03m, not paid back in 5 years
Upside: high volume, five years	−S$0.2m	S$3.6m

DBSCAN only finds where unserved searches gather; a separate demand model, on our assumptions, estimates sales. Synthetic searches. Operating profit before depreciation, interest and tax, after the contribution lost at nearby outlets. Every input and year: appendix, Expand.

MANAGER'S DECISION

Expansion manager

Which location justifies investment?

PROPOSED

Investigate the cluster near Choa Chu Kang: rerun the method on Chagee's app logs before any lease or pilot.

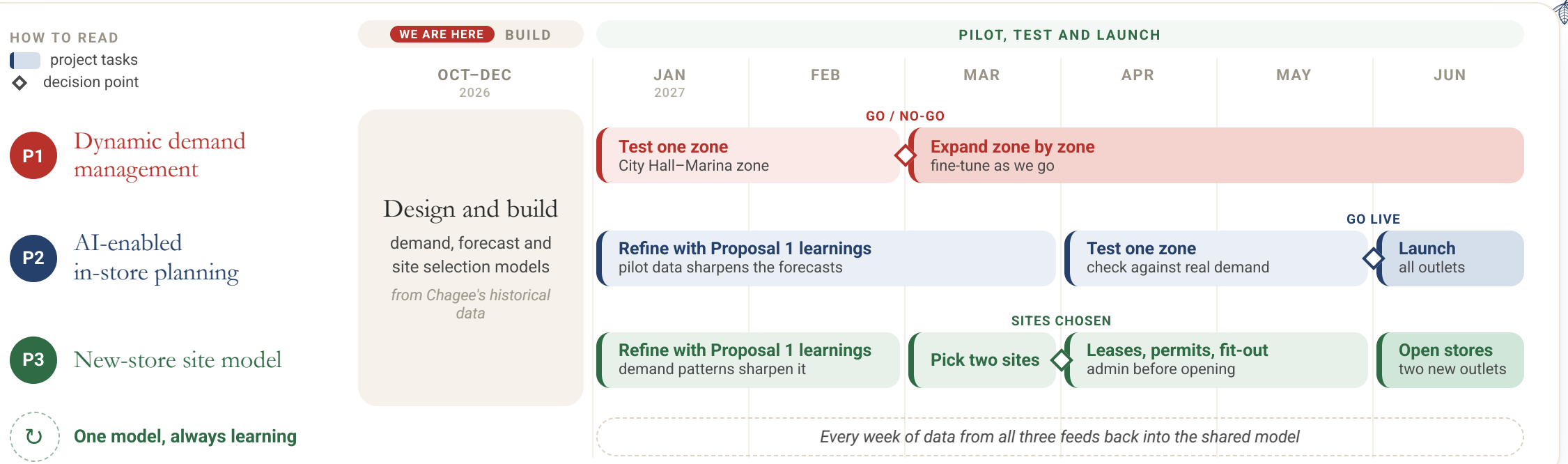
PROCEED IF, ON REAL DATA

- most of its cups come from customers without convenient access
- net additional cups, not moved cups, carry the case
- profit covers the investment, with an acceptable downside

EXECUTION PLAN

Three months to plan, then a phased six-month launch

One integrated AI solution: plan from now to December, start small in January, learn, then scale.



PROJECT TEAM — Our team: Project manager, AI engineer + From Chagee: Supply chain manager (APAC HQ), Zone manager, Store manager

By end June 2027

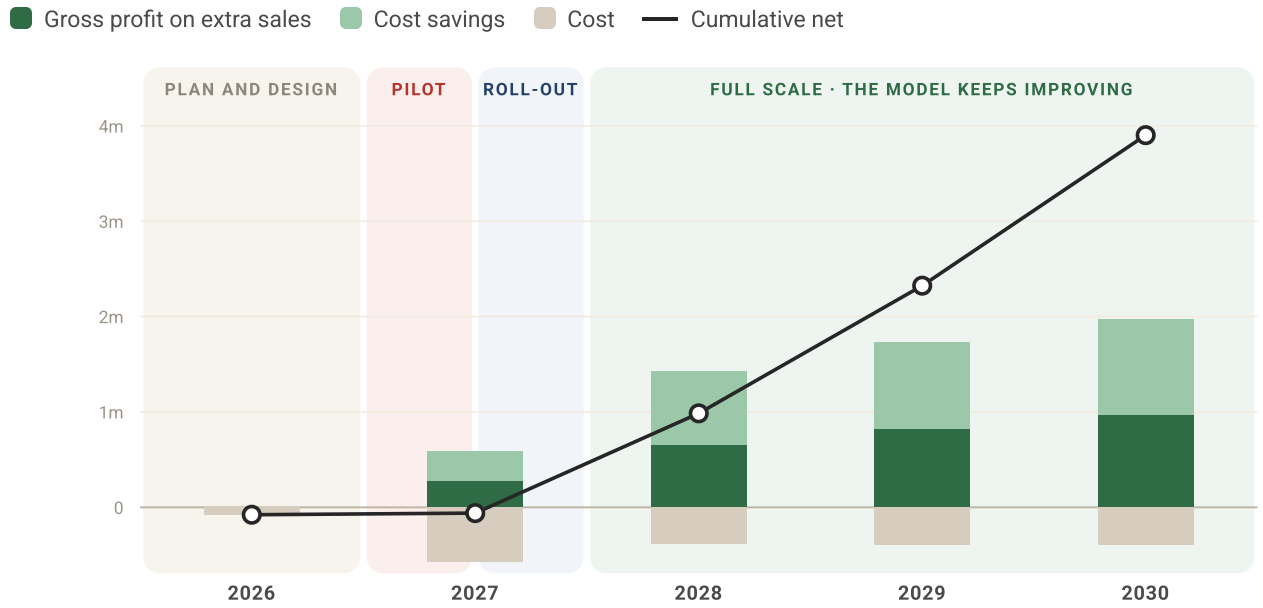
- ✓ **One integrated platform**, with Proposals 1, 2 and 3 launched across Singapore, improving demand management, decision making and site selection
- ✓ **Two new stores** opened in strong Singapore locations, chosen to maximise profit

FINANCIAL IMPACT · SINGAPORE · FIVE YEARS

We forecast our business plan to add S$3.9m net profit over five years and break even in early 2028

Team set up in 2026, pilots in early 2027, expands outlet coverage from 2H-2027 onwards; then the model keeps improving.

Financial gain and cost each year



	2026	2027	2028	2029	2030
Gain	–	+S$0.6m	+S$1.4m	+S$1.7m	+S$2.0m
Cost	−S$78k	−S$0.6m	−S$0.4m	−S$0.4m	−S$0.4m
Cumulative	−S$78k	−S$62k	+S$1.0m	+S$2.3m	+S$3.9m

REVENUE IMPACT

+S$5.0m extra revenue over five years

About 305,000 more cups a year by 2030

P1	Fewer walk-aways at peak	S$2.1m
P2	Staff and machines ready for the peak	S$1.1m
P3	Better-placed new stores	S$1.8m

COST IMPACT

S$3.0m saved

Less idle staff time, fresh-stock waste and machines

S$1.8m invested

S$0.6m one-off, then S$0.4m a year

FINANCIAL RESULT

S$3.9m net over five years

Breakeven **early 2028** · **S$1.6m** a year at full scale

Full financial model →

BEYOND THE NUMBERS — Brand value · A better customer experience · Calmer, better-looking stores

Harder to measure, just as important

Q&A

THANK YOU · QUESTIONS AND DISCUSSION

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Appendix: the detail behind each chapter

Derivations, tables, sources and earlier analyses. Click a line to open it; use the agenda to return.

EXECUTION AND IMPACT

- Financial impact: five-year model
- Financial impact: Chagee Singapore forecast
- Proposal 1 pilot: zone pilot and learning loop
- Proposal 1: the pickup race (earlier slide)
- Proposal 2 pilot: eight-week zone pilot
- Proposal 2 financial case
- Execution plan: twelve-month detail
- Proceed-if criteria by proposal
- Business impact: counter model (earlier slide)

SERVE DEMAND

- Forecast: how many people (earlier slide)
- The operations manager's decision (earlier slide)
- Bucket brigade: sharing the counter
- Automation: which steps a machine takes
- Counter comparisons, every day type
- The modelled decisions in detail
- The benefit in detail
- Machine ladder
- Machine ladder: tiers and sources

OPTIONAL EXTENSION: AUSTRALIA

- A new market without app history
- New market: inputs and method

EARLIER MAIN SLIDES

- Chagee in Singapore
- From farm to cup
- The data trail
- Our scope

EXPAND ACCESS

- How the site forecast works
- The new store in money: five-year P&L
- Four scores for every location
- The supervised model
- From pin to net profit

GROW DEMAND

- Members: when does joining feel worth it?
- Group orders as demand data
- The squad experience
- The CRM manager's squad decision
- Members: the model and its checks
- Members: break-even, assumptions and sources
- Group orders to forecasts: method and results

ACROSS THE STORY

- Chagee's data collection
- Network effects across the three parts
- Sources and assumptions

The five-year model behind the financial impact

S$ THOUSANDS UNLESS STATED	2026	2027	2028	2029	2030	TOTAL
REVENUE IMPACT						
Proposal 1 · cups recovered from walk-aways (thousands)	–	42	91	107	118	357
Proposal 2 · extra cups served at the peak (thousands)	–	20	50	59	66	195
Proposal 3 · extra cups at better-chosen sites (thousands)	–	25	64	92	122	302
Additional revenue	**–**	**513**	**1,207**	**1,519**	**1,799**	**5,038**
Less ingredients and packaging	–	(236)	(555)	(699)	(828)	(2,317)
Gross profit from additional sales	**–**	**277**	**652**	**820**	**972**	**2,720**
COST IMPACT · LEANER OPERATIONS (PROPOSAL 2)						
Staff overhead saved	–	130	328	386	427	1,272
Fresh stock waste avoided	–	141	356	419	464	1,380
Tea machines avoided	–	37	91	103	111	342
Operating cost savings	**–**	**307**	**775**	**908**	**1,002**	**2,993**
Total benefit	**–**	**584**	**1,427**	**1,729**	**1,974**	**5,714**
ADDITIONAL COST FROM INITIATIVES						
Project team	78	312	–	–	–	390
Platform build	–	100	–	–	–	100
Data pipelines and cloud set-up	–	60	–	–	–	60
Security and PDPA review	–	20	–	–	–	20
Run team	–	–	240	240	240	720
Software maintenance	–	–	20	20	20	60
Cloud, licences and AI services	–	52	70	70	70	262
Switch perks (Proposal 1)	–	23	50	59	66	198
Total cost	**78**	**568**	**380**	**389**	**396**	**1,811**
Net financial impact	**(78)**	**16**	**1,047**	**1,340**	**1,578**	**3,903**
Cumulative net impact	*(78)*	*(62)*	*985*	*2,324*	*3,903*	

Key assumptions

Some based on Chagee's real online data; the rest are illustrative assumptions

BASELINE

44 outlets × 240,000 cups a year at S$5.90, growing 3% a year, plus the new outlets: the base the percentages below apply to. Gross margin 54%; wages rise 3% a year

ROLL-OUT

Pilots in the first half of 2027; every outlet from the second half. The model reaches its full effect by 2030

PROPOSAL 1

3% of cups lost at peak, 30% of them recovered; 2% of orders redirected, 50% of them with a S$0.50 perk

PROPOSAL 2

10 idle staff hours a week saved per outlet; fresh-stock waste of 3% of sales cut by 20%; 0.5% more cups at the peak

PROPOSAL 3

Two new outlets in 2027, then one a year, each selling 10% more than a conventional site pick

COSTS

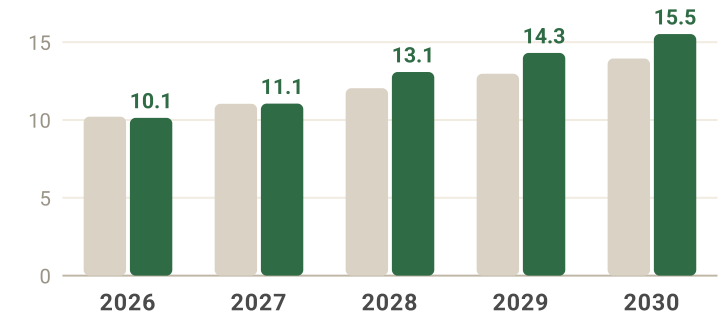
Platform build S$100k, data set-up S$60k and security review S$20k in 2027; then a run team S$240k, cloud S$70k and maintenance at 20% of the build a year. Chagee's managers and store teams are existing staff: no extra cost

With the proposals, Chagee Singapore profit grows to S$15.5m by 2030 with a 19.6% margin

The whole Singapore business, growing 3% a year with new outlets (49 by 2030), with and without the proposals.

S$ MILLIONS UNLESS STATED	2026	2027	2028	2029	2030	TOTAL
Outlets at year end	44	46	47	48	49	
Cups sold (millions)	10.6	11.2	12.0	12.7	13.4	59.9
REVENUE						
Baseline sales	62.3	65.6	69.9	73.5	77.3	348.6
Additional sales from the proposals	–	0.5	1.2	1.5	1.8	5.0
Total revenue	**62.3**	**66.1**	**71.1**	**75.0**	**79.1**	**353.6**
Cost of goods sold	(28.7)	(30.3)	(32.3)	(34.1)	(35.9)	(161.3)
Gross profit	**33.6**	**35.9**	**38.7**	**40.9**	**43.2**	**192.3**
Gross margin	*54.0%*	*54.2%*	*54.5%*	*54.6%*	*54.6%*	
OPERATING EXPENSES						
Staff	(8.6)	(8.9)	(9.3)	(9.7)	(10.2)	(46.8)
Rent and utilities	(9.2)	(9.4)	(9.8)	(10.0)	(10.2)	(48.6)
Delivery-app commissions	(5.6)	(5.9)	(6.3)	(6.6)	(7.0)	(31.4)
Equipment: tea machines avoided	–	0.0	0.1	0.1	0.1	0.3
AI platform and team	(0.1)	(0.6)	(0.4)	(0.4)	(0.4)	(1.8)
Total operating expenses	**(23.5)**	**(24.8)**	**(25.6)**	**(26.6)**	**(27.7)**	**(128.2)**
Outlet operating profit	**10.1**	**11.1**	**13.1**	**14.3**	**15.5**	**64.1**
Operating margin	*16.3%*	*16.7%*	*18.4%*	*19.1%*	*19.6%*	

FOR COMPARISON · WITHOUT THE PROPOSALS						
Outlet operating profit without the proposals	10.2	11.0	12.0	13.0	13.9	60.2
Operating margin without the proposals	*16.4%*	*16.8%*	*17.2%*	*17.6%*	*18.0%*	
Uplift from the proposals	**(0.1)**	**0.0**	**1.0**	**1.3**	**1.6**	**3.9**



REVENUE IN 2030

S$79.1m

up 27% on 2026, with 49 outlets

MARGIN IN 2030

19.6%

against 18.0% without the proposals

Pilot smart pickup routing in one zone, then expand on evidence

A zone is a group of reachable outlets. Start with one busy outlet and a nearby support outlet. Back to how it works →

Zone pilot

Suntec City
busy origin outlet

⇄

Raffles City
support when capacity permits

A zone is a group of reachable outlets; an outlet can change roles as demand shifts.

Normal checkout → **Routing alone** → **Routing + bag offer**

Measure, assess, refine

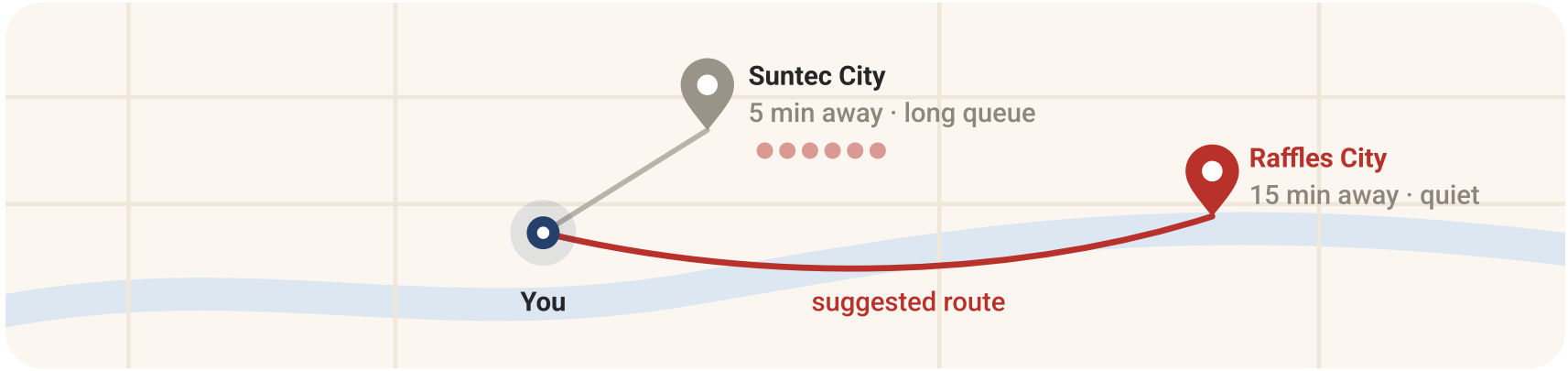
1. **Measure** switches, actual collection times, both queues and zone contribution
2. **Assess** the traffic shift: generative AI summarises results and flags congestion moving
3. **Refine and expand**: managers approve policy changes and new zones

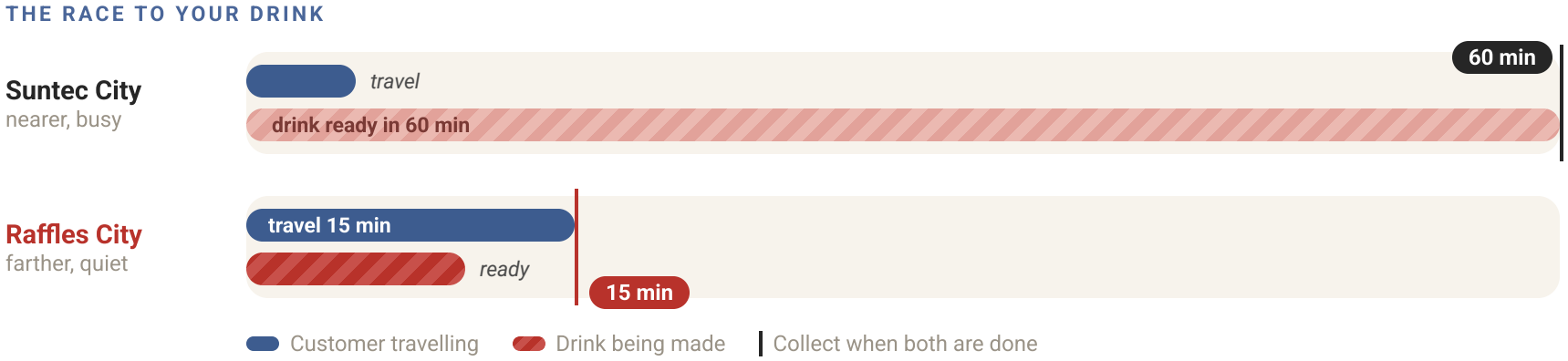
ROLLOUT GATE

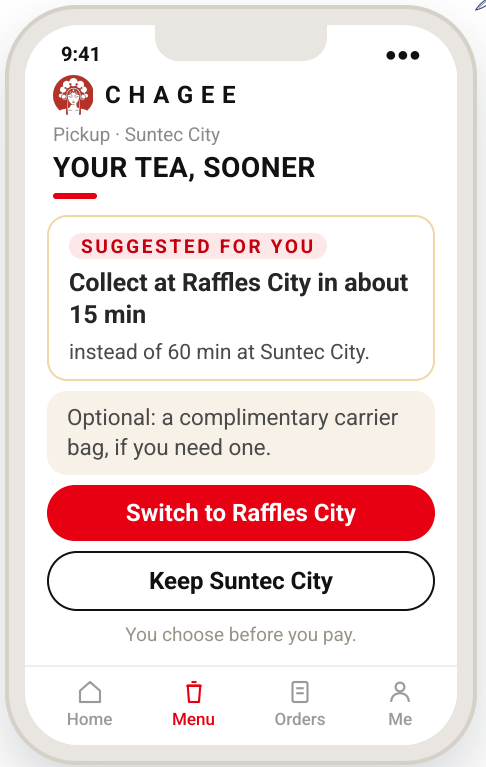
Faster collection, reliable support outlets and positive net contribution across the zone.

Your tea, sooner: AI routes pickups to a faster outlet

When one outlet is swamped, the app offers a nearby outlet that can hand over the drink sooner, and the customer chooses.







GENERATIVE AI MESSAGE · MOCK-UP

45 minutes sooner

Travel and preparation happen at the same time, so the customer collects as soon as both are done.

An eight-week zone pilot that separates demand from constraints

One outlet can test forecasting and rosters; a zone is needed to test staff moves and their effect on the network.

Back to how it works →

EIGHT-WEEK ZONE PILOT

W1–2 Baseline and shadow forecasts

Choose one busy outlet and one or two support outlets. Audit the data and today's planning.

W3–6 Test rosters and staff moves

Compare matched periods. Include travel time, cover at the lending outlet and holiday demand.

W7–8 Review, diagnose and decide

Check the benefit across the whole zone. Extend if holiday or peak data is too thin.

OWNERS regional operations · outlet managers · data team · finance

IF ONE OUTLET TAKES ABOUT THREE TIMES LONGER

Investigate before adding capacity

Before preparation

Arrival surges, staffing, how orders are released

During preparation

Machine queues, downtime, layout, restocking

After drinks are ready

Hand-over, and customer or rider arrival

The financial case depends on realised improvements

Measure avoided paid hours, retained sales and equipment returns across the whole zone. Back to how it works →

ILLUSTRATIVE MONTHLY ZONE CASE

Benefit or cost	S$ a month
Avoided overtime: 60 h × S$15.06	904
Recurring system and support cost	−300
Net, before set-up and sales upside	**604**

20 paid hours a month

cover the S$300 recurring cost. Set-up, training and transfer costs are extra.

WHAT FINANCE MUST VERIFY

Labour savings

Count only fewer paid hours or less overtime. Moving an employee is not itself a saving.

Retained sales

Count otherwise-lost purchases at contribution margin, net of added costs.

Waste

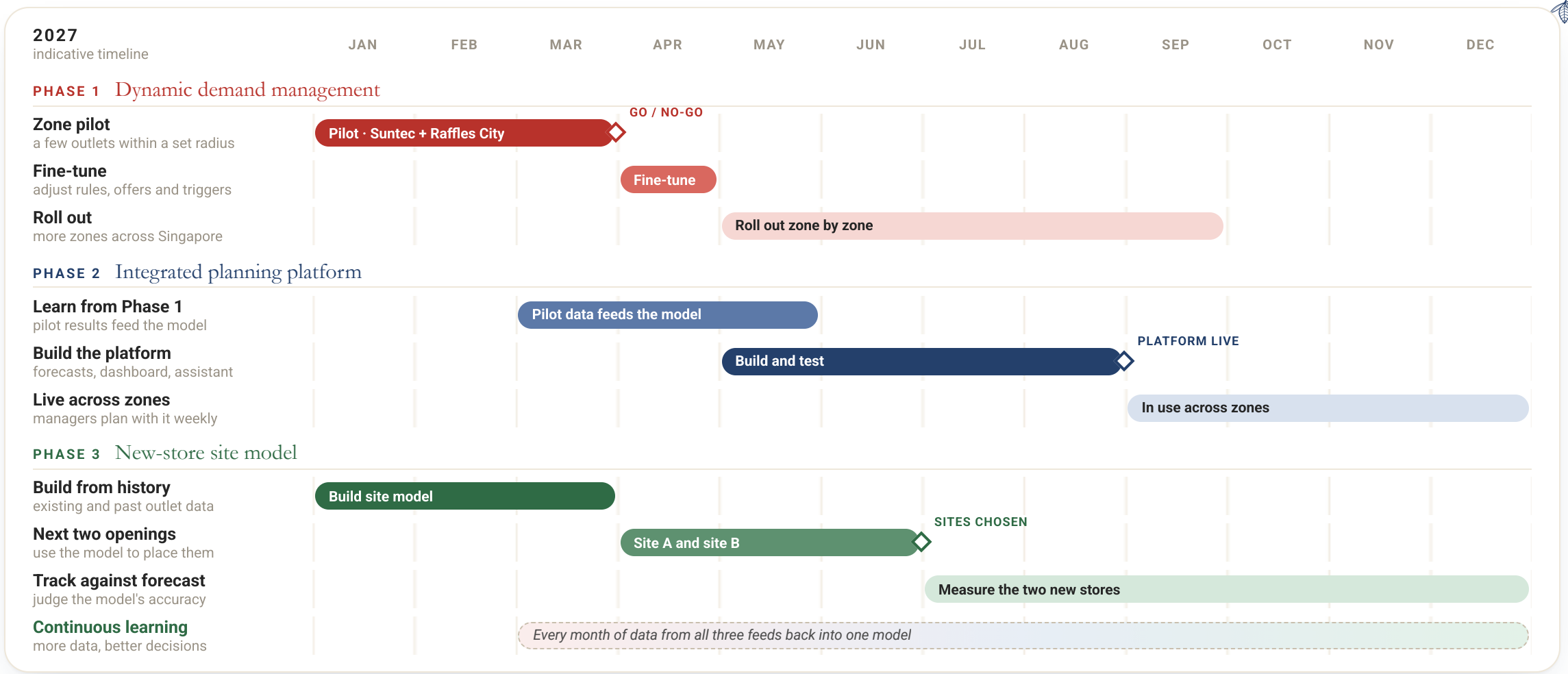
Count only lower write-offs of perishables, not stock moved between outlets.

Equipment

Approve only if the extra cash benefit justifies the installed cost and running expense.

One integrated AI solution, delivered in three phases

Start small, learn, then scale. Each phase feeds the next, and the model keeps improving as data grows.



Phase 1 · Phase 2 · Phase 3 · ◇ decision gate

Proposal 1 pilot → · Proposal 2 pilot →

Three managers, three pilots, each judged on its own evidence

A connected story, not one combined model: no simulated benefit is carried between parts or added up.

PROPOSAL 1 · SMART PICKUP ROUTING

Which zone should we pilot routing in?

RECOMMEND

Pilot in one zone, a busy outlet with a nearby support outlet: normal checkout, then routing alone, then routing with the bag offer.

PROCEED IF

collection is faster, support outlets stay reliable, and net contribution across the zone is positive.

PROPOSAL 2 · AI-ENABLED IN-STORE PLANNING

Which zone should pilot AI in-store planning?

RECOMMEND

An eight-week zone pilot: one busy outlet and one or two support outlets, testing AI rosters, staff moves, machine and stock plans.

PROCEED IF

paid hours and waste fall, waits hold or improve, and net benefit across the zone is positive.

EXPAND ACCESS · EXPANSION MANAGER

Which location justifies investment?

RECOMMEND

Investigate the cluster near Choa Chu Kang on Chagee's real app logs before any lease.

PROCEED IF

it adds net new customers, and its profit covers the investment with an acceptable downside.

TAKEAWAY | **Get customers their tea sooner → serve demand reliably → open where we add new customers.**

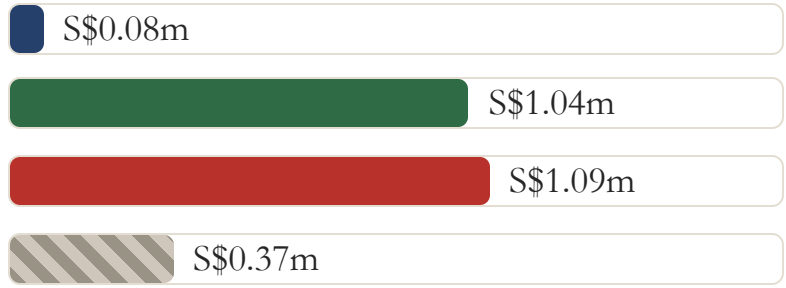
 TBC · DRAFT, TO BE CONFIRMED

About S$1.09m a year across 44 outlets, mostly from one machine

In our model, against an illustrative baseline: modelled avoided costs, not demonstrated savings.

PROFIT A YEAR, ALL 44 OUTLETS · ECONOMIC RULE THROUGHOUT

AGAINST THE ILLUSTRATIVE ALL-HAND BASELINE (FLAT NORMAL-DAY ROSTER)



PILOT COMPARISON: THE PROPOSED SCENARIO AGAINST TODAY'S SPLIT (OUR SCENARIO), BOTH ON THE ECONOMIC RULE



Profit = orders kept × price × margin, plus labour saved, minus machine cost. Calendar: 104 busy days (weekends) and 11 event days (public holidays) a year, an illustrative definition. Baseline: all by hand, the normal-day crew every day. Peak hour only, so orders kept are a floor. Every figure, range and both crew rules: appendix.

PHYSICAL AI · ONE TEA-AND-MILK MACHINE

Most of the benefit: labour

Lets the roster drop from 2 people to 1 in the off-peak hours: **S$1.04m** a year after the machine's cost.

FORECAST + ECONOMIC RULE

Adds event-day orders on top of the machine

Adds a person on event days only (2 → 3), keeping about 35 more orders per outlet that day: 390 a year, **S$0.05m** across Singapore. On its own, all by hand: S$0.08m.

READ WITH CARE

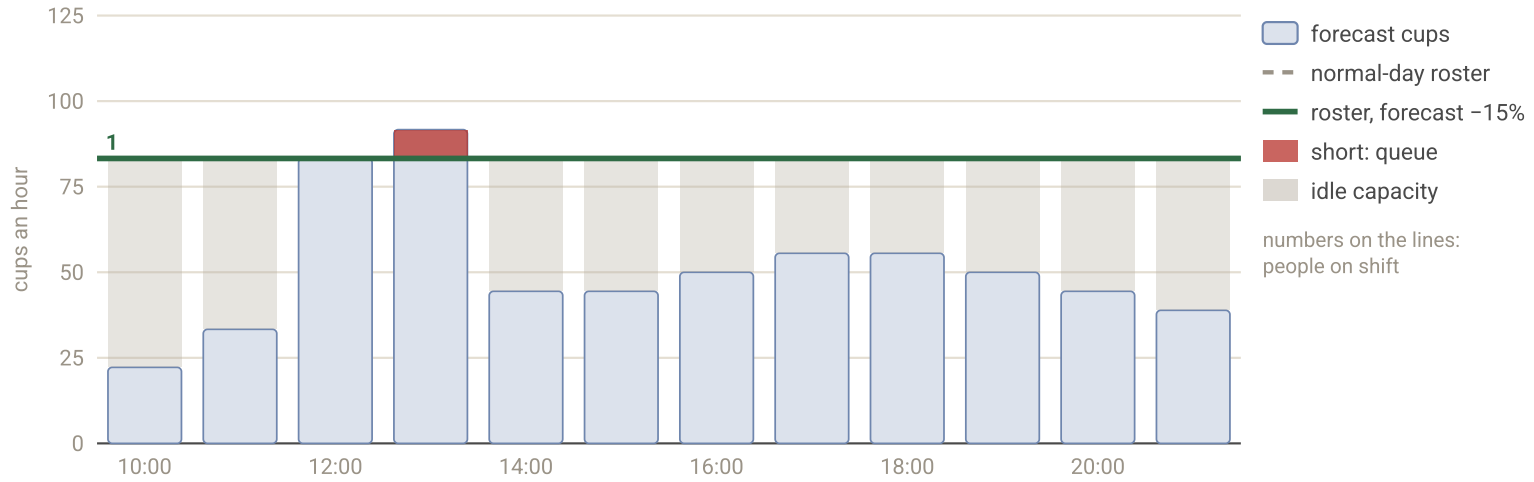
The big number rests on one assumption

The minimum crew, 2 by hand and 1 with machines, is ours. Without it, S$1.09m falls to **S$0.37m**. The pilot arm beats today's split by S$1.22m: more than the headline, because today's split is itself S$0.14m below the baseline.

A forecast sets the crew each hour needs

BUSINESS PROBLEM Demand changes by the hour; staffing and fixed station assignments may not, leaving queues, walk-aways or idle capacity. Observed people deployed on local expertise (Chagee source, anonymous); long reopening queues (Vulcan Post). Simulated everything below. To confirm Chagee's waits and walk-aways.

CUPS AN HOUR AGAINST CREW CAPACITY · BUSY DAY · ILLUSTRATIVE SIMULATION



SHOWN
Busy day, today's split, forecast −15% low. Other days: the deck's interactive version.

TAKEAWAY | **Forecast −15% low: 1 at the peak, 20.8% walk away, S$372 a day; peak-hour contribution S$233 against S$287 with an exact forecast.**

Roster to the forecast, test the brigade, pilot one machine

ONE LEVER PER COLUMN · PEAK HOUR · BUCKET BRIGADE · EVENT DAY · MEAN OF 20 RUNS

CRITERION	TODAY normal-day roster · today's split	+ FORECAST forecast roster · today's split	PROPOSED PILOT + ONE MACHINE forecast roster · tea-and-milk machine
Crew at the peak	1	2	**3**
Median wait	7:24	2:38	**1:44**
Walk-aways	53.8%	1.2%	**0.2%**
Cups served per person, peak hour	69	74	**50**
Peak-hour contribution	S$230	S$491	**S$482**
Cost a day: roster and machines	S$372	S$402	**S$335**
Walk-aways with fixed posts: well / badly assigned	54.2% / 54.2%	4.1% / 34.9%	**0.4% / 22.4%**

READING THE TABLE

Green is the proposed pilot. Each column adds one lever: the forecast changes only the crew, the machine only the equipment. The last row swaps the brigade for fixed posts. Illustrative simulation.

IF THE FORECAST IS OFF

A busy-day forecast 15% too low leaves about **21%** walking away (the forecast slide).

MANAGER'S DECISION

Operations manager

What crew and counter setup should we use?

PROPOSED

Pilot in 3 outlets: shadow the forecast, roster to it, then run one tea-and-milk machine against today's split.

PROCEED IF

- waits within 5 min, walk-aways within 2%
- cups per labour hour and contribution at least today's
- it holds when the forecast is off

NEXT Serving customers better also means being where demand is still unserved.

Company-owned outlets, one data system, one counter

HOW A SINGAPORE CUP IS SOLD

SINGAPORE: COMPANY-OWNED STORES

SG **Chagee runs its Singapore outlets itself**, not through franchisees, so the counter design, roster and equipment are Chagee's own decisions (confirmed by the project owner).

Global context only: group-wide, most FY 2025 net revenue came from franchising (88%, secondary source). That does not describe Singapore.

SG Singapore outlets run under one company and one data system (Chagee source, anonymous)

In store	Walk-in (app order; tourists can order at the counter, to confirm)
Chagee app	order ahead, collect
Delivery	third-party delivery apps (e.g. Grab); no Chagee-run delivery

→ **One counter** every channel's cup is made here



44 outlets, ground-floor mall units by preference; kiosks in basements and MRT stations at entry in 2019

- **Chagee-App**

  An **order-ahead, self-pickup app**; in Singapore, **in-app ordering and delivery** followed the reopening queues. Chagee **collects personal and transactional data**.

  To confirm: the app's share of Singapore orders; whether searches or location are logged.

  · CHAGEE App Store listing · Vulcan Post, Oct 2024
  · Chagee Form 20-F, FY 2025

- **Why Chagee?**

  1. **One company, one data system** (Chagee source, anonymous), with **lots of data** from its reliance on the Chagee app.
     Data collection: appendix →
  2. A **fast-growing, playful** brand with **heavy app use**. Likely to have **rich order data** and **room for AI** to enhance the business.
  3. We also **interviewed an internal staff member** (anonymous) about the problems they actually face.

TAKEAWAY | **Whatever the channel, every order ends at one counter. Peak hour is where an outlet wins or loses.**

Supply is planned ahead; the counter feels demand by the minute

FOLLOW ONE CUP · STEP THROUGH

Source

Tea leaves from **Yunnan** through standard channels for every market

DATA CREATED

Shipments, lead times

Local: fresh milk, fruit

Warehouse

Separate central and regional warehouses keep storage and logistics costs down

DATA CREATED

Stock vs par, expiry

Outlet Counter

Machines: brew each cup from a scanned recipe
People: Ice, toppings and hand-over

DATA CREATED

Cups by hour

Wait times (to confirm)

Last Mile

Self-collect via app, or delivery via **third-party** apps such as Grab. **Riders** wait at the same counter

DATA CREATED

Orders by channel

Rider wait (to confirm)

Customer

Orders through app and **collects** at the counter.

DATA CREATED

App orders, drop-offs

TAKEAWAY | Chagee already plans its stock (public sources). People are deployed on local expertise, by the GM and outlet teams (Chagee source, anonymous); that is the call we model.

Orders are recorded; the other forecast inputs are still to confirm

WHICH SOURCE FEEDS WHICH DECISION · STEP THROUGH

	Store and production	App and members	Delivery	Supply and context
Demand forecasts	Cups per hour by drink and channel	Member repeat demand	Orders per hour by platform	Weather, holidays, launches
Staffing and prep	Wait times, machine logs	Pickup no-shows	Hand-over peaks	Labour hours vs cups per hour
Replenishment	Drink mix per outlet	Launch trial rates	Group-order toppings	Ingredient use, expiry, par
Queue and capacity	Wait time, footfall vs sales	Store-select drop-offs	Rider wait at counter	Stock-outs at peak
Site selection	Sales density per outlet	"No store nearby", home and work	Distance and fee by postcode	Mall footfall, competitors
Menu and promos	Drink share, customisation	Trial and repeat, voucher uplift	Delivery basket mix	Collab calendar

White chips: core input; grey: refines it; red rows: our scope. Wait times, footfall, searches and location are to confirm.

TAKEAWAY | Orders by hour are recorded; searches, location, waits and walk-aways need confirming. Today, GM and outlet teams deploy people on local expertise (Chagee source, anonymous); we show how a forecast could support those calls.

Grow demand, serve it well, then expand access

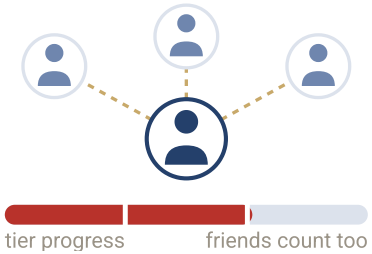
Three connected decisions, three owners. Each chapter: business problem → demonstration → manager's decision.

THREE CONNECTED DECISIONS

1

Grow demand

Make membership more rewarding.



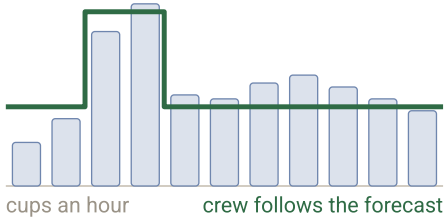
CRM MANAGER
Which features and rewards should we pilot?

Squads: appendix

2

Serve demand

Match staffing and workflow to demand.



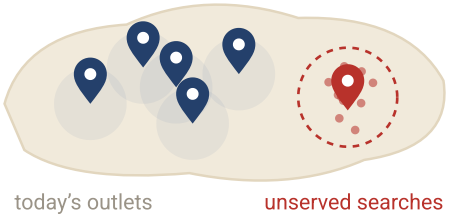
OPERATIONS MANAGER
What crew and counter setup should we use?

Slides 08 and 09

3

Expand access

Open where we add new customers.



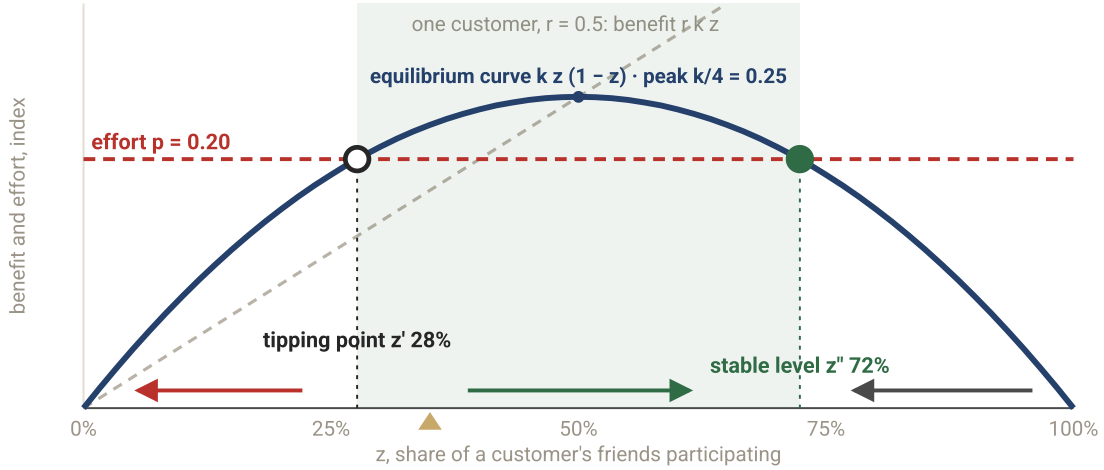
EXPANSION MANAGER
Which location justifies investment?

Slides 10 and 11

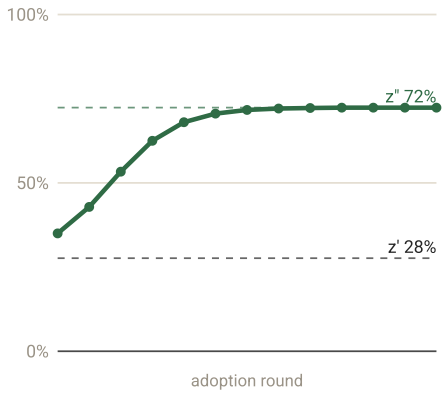
Squads add value only once enough friends take part

BUSINESS PROBLEM Chagee's Membership Operations team told us the problem they want to solve is **moving customers up the membership tiers**: customers may not think the effort of progressing is worthwhile.

BENEFIT AGAINST EFFORT · NETWORK-EFFECTS MODEL, NOT AI · ILLUSTRATIVE INDEX



PARTICIPATION, ROUND BY ROUND

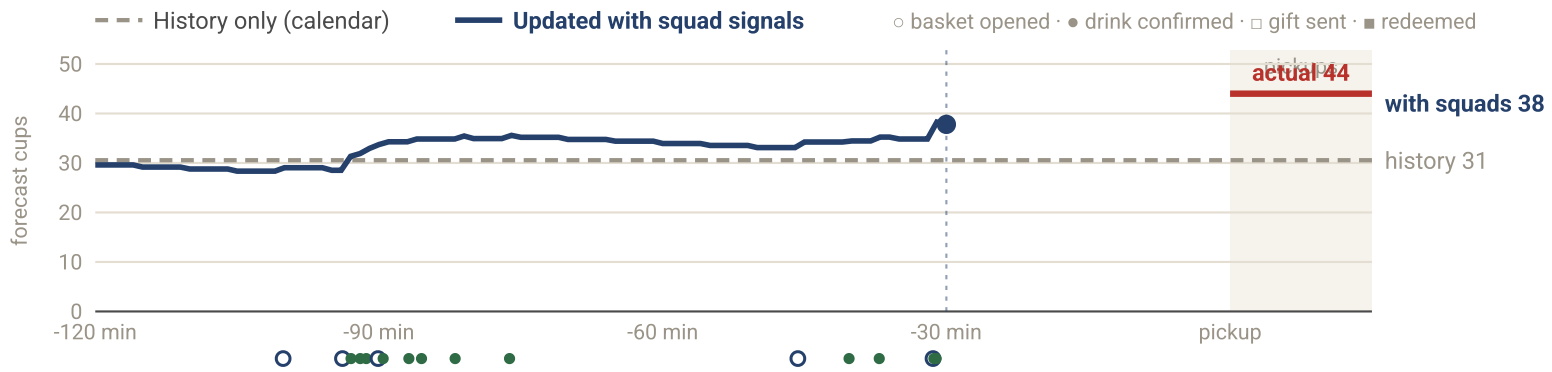


SHOWN
Effort score 0.20 (an illustrative index, not money), value of friends k 1.00, starting participation 35%.

TAKEAWAY | **Tipping point 28%, stable level 72%. Starting at 35%, participation rises to about 72%. Illustrative.**

Every group order helps the outlet prepare for demand

CUPS FOR THE 13:00 TO 13:30 PICKUPS · SYNTHETIC



Predict

38 cups expected at -30 min =

21 usual, from history + **13** confirmed in squads + **3.4** likely

Likely: invites not yet confirmed and gifts not yet redeemed, each counted at the chance it converted in past windows (21% and 19%), plus baskets not yet opened.

Act

Prepare
Jasmine 14 · Oolong 13 · Black 6
drinks to have ready

Staff
1 person, 13:00 to 13:30
the window's crew

Counter
one person, machines on tea and milk · load 76%
how the line is run

Pickup estimate
ready in about 3 min
shown to customers (rough)

SHOWN

Squad rush (synthetic), squad share 30%, forecast at -30 min, actual revealed. Window 13:00 to 13:30.

HOW TO READ IT

1. **Grey dashed**: what history alone predicts for these pickups.
2. **Navy**: the same forecast, updated as squad orders come in (dots under the chart).
3. **Act**: the outlet prepares, rosters and quotes pickup times from the navy number.
4. **Reveal actual**: see which forecast was closer.

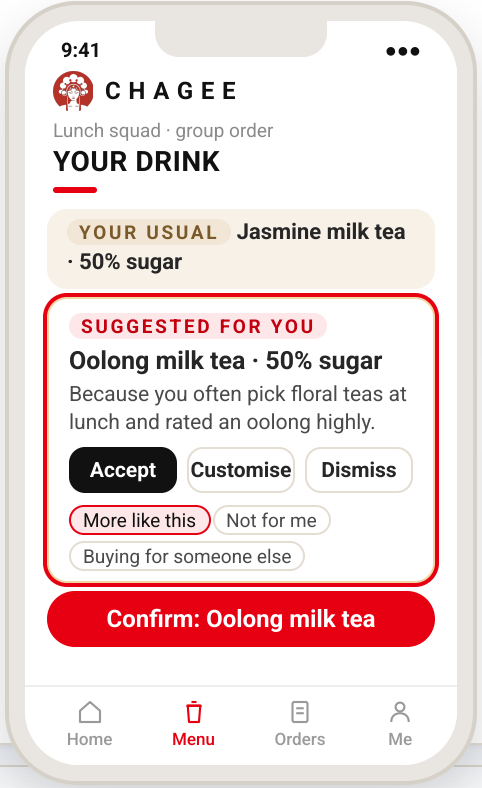
TAKEAWAY | **Squad rush: actual 44 cups. At -30 min the history-only forecast was off by 13.4, the squad forecast by 6.2: squad orders showed the rush coming before history could. Across 300 synthetic windows the squad forecast was closer on 67% of them.**

In the app, friends' orders count for everyone

BUSINESS PROBLEM Chagee's Membership Operations team told us the problem they want to solve is **moving customers up the membership tiers**: customers may not think the effort of progressing is worthwhile.

PROPOSED EXPERIENCE · ILLUSTRATIVE PROTOTYPE

1. See your progress and start a squad
2. Invite friends from the share sheet
3. Choose your drink: your usual, or a suggestion
4. Each friend confirms their own drink
5. Collect the order: everyone's progress moves
6. Send a friend a drink
7. The friend picks from suggestions made for them



BASELINE AND AI
Recalling your last order is the baseline. The proposed AI part ranks other drinks from your own choices, acceptances and dismissals. Your feedback teaches it; "Buying for someone else" keeps that drink out of your profile.

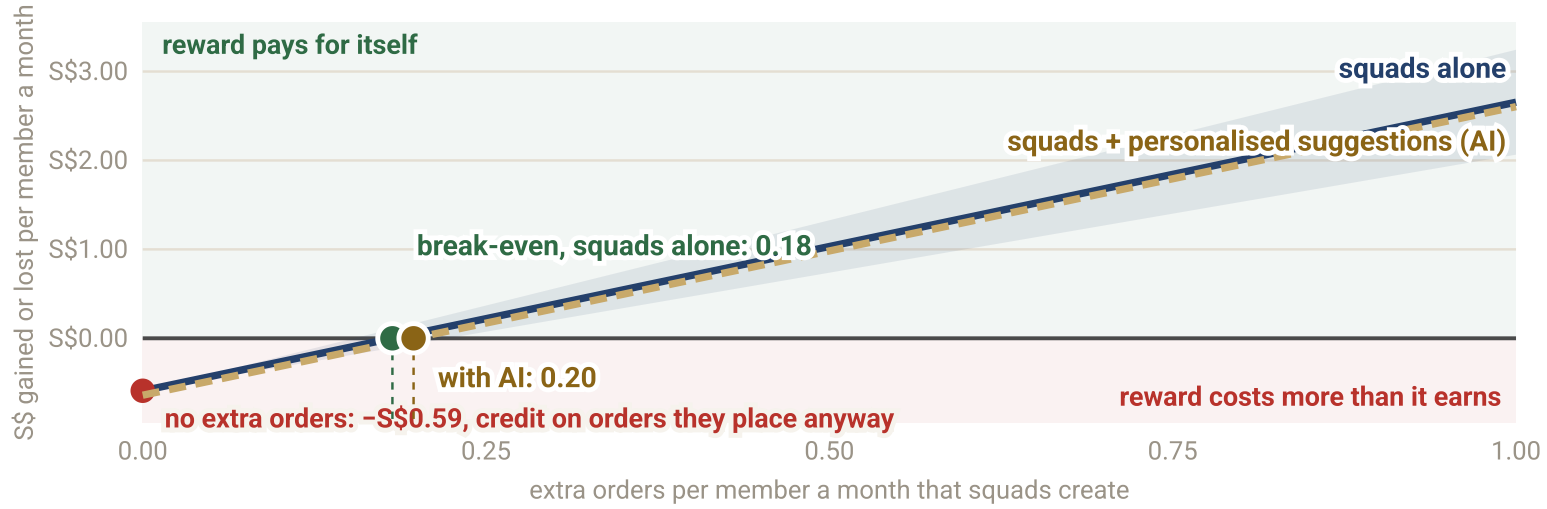
WHY IT MATTERS
Easier choices may lower the effort, p, further. A hypothesis to test, like k.

Proposed experience with illustrative data and rules, not Chagee's; nothing is learned from real customers.

TAKEAWAY | **At the modelled effort of 0.20, squads (assumed k 1.00 → 1.30) would move the tipping point from 28% to 19%: fewer friends need to be on before joining feels worth it. Assumptions to test, not measured effects.**

Pilot squads only at a reward their extra orders can pay for

DOES THE REWARD PAY FOR ITSELF? · AFTER REWARD AND AI COSTS · ILLUSTRATIVE



SHOWN

Credit 5.0% of the S$5.90 price; 2.0 group orders a month anyway; AI cost S$0.05 a member a month; margin 60% (band 50%–70%).

HOW TO READ IT

1. **Start**: no extra orders costs **S$0.59** a member a month, the credit on orders placed anyway.
2. **Rise**: each extra order earns its margin (band: 50%–70%).
3. **Break-even**: **0.18** extra orders a member a month, about one every 5 months.
4. **AI** adds S$0.05 a month, so **0.02** more.

MANAGER'S DECISION

CRM manager

Which features and rewards should we pilot?

PROPOSED

Three arms in Singapore (today's membership, squads, squads with suggestions) at a **5.0%** credit: squads must add **0.18** orders a member a month, suggestions **0.02** more.

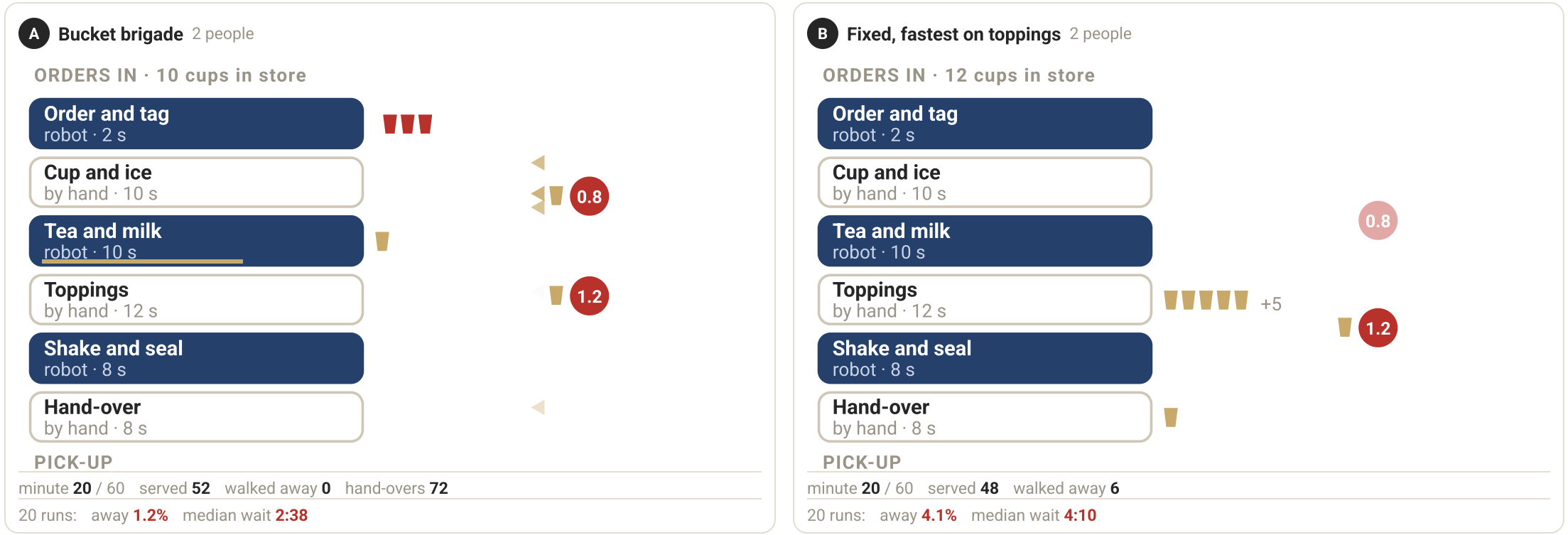
PROCEED IF, ARM AGAINST ARM

- tier progression and orders rise against today's membership
- satisfaction holds
- contribution after reward and AI costs is positive

NEXT A more engaging membership must be matched by reliable fulfilment.

A bucket brigade balances a lean crew by itself

PEAK HOUR · SAME CUSTOMERS ON BOTH SIDES · ILLUSTRATIVE SIMULATION



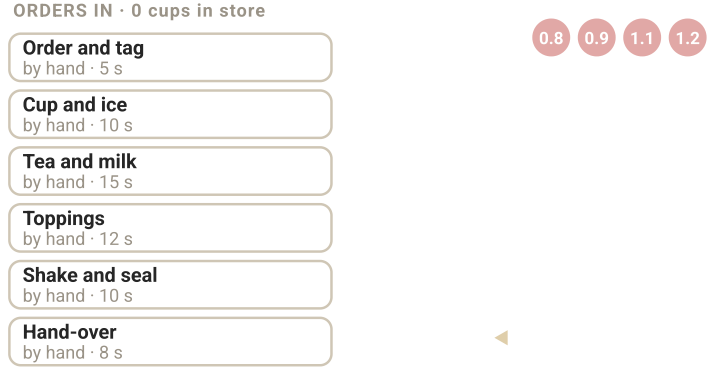
Red circle = a person (number = speed) · gold ring = held back · faded = walking · ▲ = hand-over · navy = robot · red cup = waiting for a person

TAKEAWAY | With a lean crew, work flows to whoever is free, so the line balances itself. With spare people, well-assigned fixed posts are as good or a little faster.

In our model, one tea-and-milk machine is cheapest every day

PEAK HOUR · ILLUSTRATIVE SIMULATION

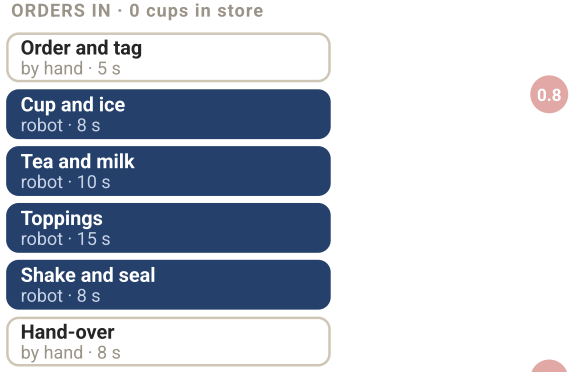
A All by hand 4 people, brigade



minute **20** / 60 served **57** walked away **0** hand-overs **156**

20 runs: away **0.0%** wait **1:25** **S$422**/day

B Machine-made drink (scenario) 2 people, fixed posts



minute **20** / 60 served **57** walked away **0**

20 runs: away **0.0%** wait **1:06** **S$465**/day

Navy = robot, white = by hand (click to switch) · red circle = a person · faded = walking · red cup = waiting for a person.

COST A DAY · EVENT DAY · WITHIN THE WALK-AWAY LIMIT

One machine (tea and milk) ✓	**S$335**
Today's split (our scenario)	**S$402**
All by hand	**S$422**
Machine-made drink (scenario)	**S$465**
All robots	**S$562**

✓ cheapest that keeps walk-aways within 2%. Full table: Numbers.

TAKEAWAY | One machine on tea and milk lets the roster drop a person off-peak for the cost of one unit. That holds only while a unit costs more than about S$905 a month; Chagee's real cost is unknown.

OPTIONAL EXTENSION

A NEW MARKET WITHOUT APP HISTORY: AUSTRALIA

WHAT WE CAN KNOW BEFORE OPENING · WHICH AREAS · WHICH SETUP · PILOT, LEARN, THEN EXPAND

 New-market worked example · Australia · scenario assumptions

No Chagee sales history is not the same as no data: profile each area on what a new market can measure

Archetype candidate areas in one Australian state capital, standing in for real suburbs until the data is pulled. We assume Chagee has no operating history there.

CANDIDATE-AREA PROFILES · CONSISTENT FEATURES, EVERY AREA

FEATURE	STANDS IN FOR	A	B	C	D	A REAL VALUE WOULD COME FROM	VALUE HERE
Residents within 1 km	proxy walk-in demand from people living nearby	12k	18k	25k	20k	national census, usual residents by small area	assumption
Workers within 1 km	proxy weekday lunch and after-work demand	90k	20k	12k	5k	census place-of-work counts	assumption
Tertiary students within 1 km	proxy the young, app-first customer base	15k	40k	5k	2k	university and TAFE enrolment by campus	assumption
Household income index	proxy willingness to pay for a premium tea	115	80	95	90	census median household income	assumption
Transit access	proxy footfall past the counter	0.95	0.75	0.70	0.35	public-transport timetable data, services per hour at the nearest stops	assumption
Competitors within 500 m	proxy rival bubble-tea and cafe supply (lowers the score)	9	6	4	2	business listings or a map-places count	assumption
Rent, A$ a month	proxy local cost (lowers the score)	A$22,000	A$12,000	A$10,000	A$8,000	retail lease listings or an agent quote	assumption

Areas (archetypes): A CBD office core · B University precinct · C Suburban rail interchange · D Outer-suburb shopping centre

FOUR KINDS OF INPUT

- Sourced observation — none yet: no Australian data retrieved for this deck
- Proxy — what each feature stands in for, such as workers for weekday demand
- Scenario assumption — every value in the table
- Unavailable — before opening; left out, below

LEFT OUT · NEEDS CHAGEE ACTIVITY

- **Chagee app searches nearby**: needs Chagee app users in the market
- **Delivery orders from the area**: needs Chagee on the local delivery platforms
- **Distance to the nearest Chagee outlet**: there is none yet
- **Sales of existing Chagee outlets nearby**: there are none yet

Missing is unknown, not zero

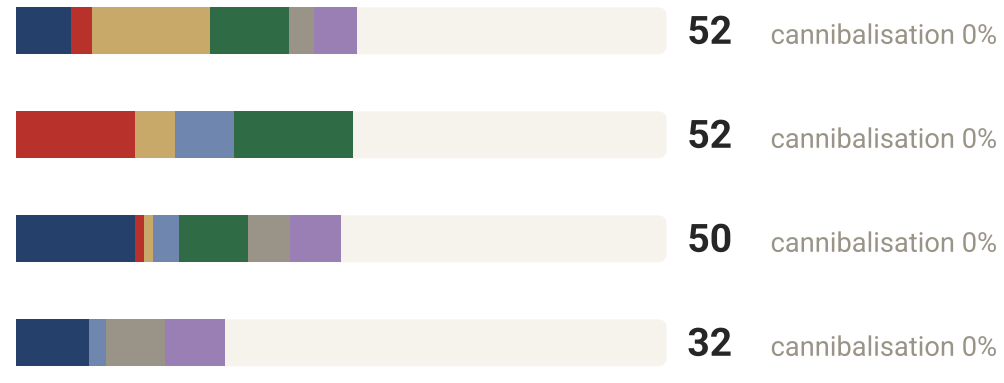
No area has app searches or delivery orders yet. Set to zero, they would read as no demand; left out, the areas are compared only on what all of them have.

WHAT IS AI HERE Nothing yet. This is the data step the models on the next slides need; every value stays a scenario assumption until census and market data are pulled.

 New-market worked example · Australia · scenario assumptions

B, A and C can each rank first as the weights move; the score shortlists areas, it does not forecast their sales

SUITABILITY SCORE OUT OF 100 · ILLUSTRATIVE WEIGHTS



B · University precinct rank 1–3 across 15 weightings · first in 6	**52**	cannibalisation 0%
A · CBD office core rank 1–3 across 15 weightings · first in 6	**52**	cannibalisation 0%
C · Suburban rail interchange rank 1–3 across 15 weightings · first in 3	**50**	cannibalisation 0%
D · Outer-suburb shopping centre rank 4 across 15 weightings · first in 0	**32**	cannibalisation 0%

Residents · Workers · Students · Income · Transit · Competitors · Rent · bar = score; rank range across the weight check

DEMAND SCENARIOS, CUPS A DAY · THE SAME FOR EVERY AREA

Low 278 Base 556 High 833

A placeholder scale: our Singapore per-outlet volume assumption, reused. With no calibration from real outlets, the score cannot turn into cups, so every area faces this range.

WEIGHTS USED

Residents 1.00 · Workers 1.00 · Students 1.00 · Income 0.50 · Transit 1.00 · Competitors 0.50 · Rent 0.50. Competitors and rent count against.

WEIGHT CHECK · 15 WEIGHTINGS

Each weight dropped, then doubled, one at a time. **B** first in 6, **A** first in 6, **C** first in 3, **D** first in 0. The top spot is not robust: treat the leaders as a shortlist.

WITH REAL OUTCOMES · SUPERVISED LEARNING

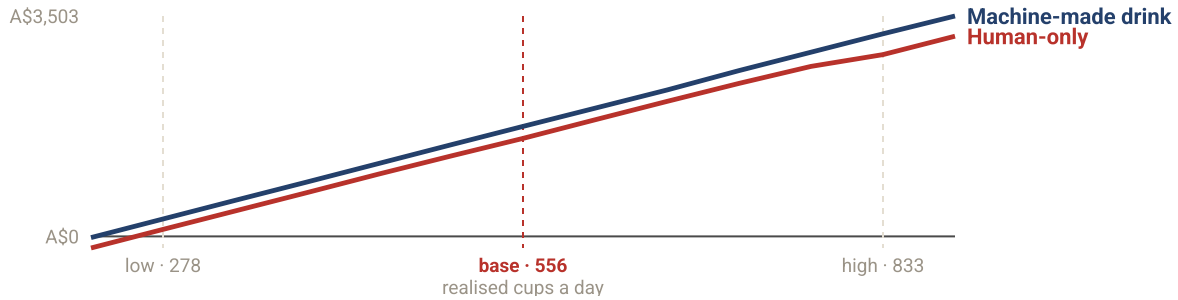
If outlets in existing markets had real sales, with these same features measured around each, a supervised model could learn the weights instead of us choosing them, tested on markets it has not seen. The supervised site model, also under Things to consider, is trained on synthetic Singapore data: **not validated for Australia**, and not used here.

Illustrative. Archetype areas and scenario assumptions: not Australian data, not Chagee data.

New-market worked example · Australia · A$ scenario assumptions

With our A$ assumptions, machine-made drink holds up best in the worst case; human-only wins if a machine unit costs more than about A$3,258 a month

OPERATING CONTRIBUTION, A$ A DAY, YEAR AVERAGE · WITH AI STAFFING, EXACT FORECAST



EVENT DAY, BASE DEMAND (556 CUPS)	PEAK CREW	MEDIAN WAIT	WALK AWAY	A$ A DAY
Human-only · no machines				
Without AI · one crew all day	2	5:33	40.1%	A$1,766
With AI · forecast −30%	3	4:28	10.2%	A$2,139
With AI · forecast exact	**4**	**1:25**	**0.0%**	**A$2,218**
With AI · forecast +30%	5	1:08	0.0%	A$2,146
Machine-made drink · 4 machine units				
Without AI · one crew all day	1	5:51	5.4%	A$2,448
With AI · forecast −30%	1	5:51	5.4%	A$2,448
With AI · forecast exact	**1**	**5:51**	**5.4%**	**A$2,448**
With AI · forecast +30%	2	1:18	0.0%	A$2,491

Machines = automation: equipment sized before opening, held fixed. **AI = AI-assisted staffing**: the forecast sets the crew; realised demand unchanged. Event day, base demand · peak hour · 20 runs · illustrative

SHOWN

Event day, base demand, forecast exact in the band. Wage A$36, machine unit A$2,000 a month, rent A$12,000 a month (University precinct).

BREAK-EVEN, CUPS A DAY · AREA B

Human-only 268 with AI, 267 without; **Machine-made drink** 236 with AI, 235 without. Normal day, no walk-aways; low scenario 278.

PREFERRED SETUP · WORST CASE

Worst year-average day, any scenario and forecast error: human-only **A$108**, machine-made drink **A$276**. **Machine-made drink** is safer; the two cross at a machine unit of about **A$3,258** a month (now A$2,000); they do not cross within the wage range. With an exact forecast, machine-made drink leads at low, base and high demand.

WITHOUT AI INTELLIGENCE
one crew, all day

Human-only: 2 people · away **40.1%** · wait 5:33 · A$1,766/day
Machine-made drink: 1 person · away **5.4%** · wait 5:51 · A$2,448/day

WITH AI INTELLIGENCE
forecast roster

Human-only (forecast exact): 4 people · away **0.0%** · wait 1:25 · A$2,218/day
· +2 at the peak, +4 staff hours, +A$452
Machine-made drink (forecast exact): 1 person · away **5.4%** · wait 5:51 · A$2,448/day
· same crew: the forecast changes nothing here

New-market worked example · Australia · scenario assumptions

Pilot one machine-made drink counter in the University precinct, then let the pilot's own data decide the next step

RECOMMENDATION · CONDITIONAL

Where: the University precinct (B)

First at our weights; B, A and C are the shortlist. No own-store cannibalisation.

Setup: machine-made drink, AI staffing

Best worst case, A$276 a day against A$108 for human-only.

Holds only if

- a machine unit costs under about A$3,258 a month
- volume lands in the placeholder range, 278 to 833 cups a day; break-even is about 236
- the A$ wage, price and rent hold
- the forecast misses by no more than 30%

WHAT THE PILOT MEASURES, AND WHAT IT UPDATES

MEASURE	HOW	UPDATES
Orders, cups a day	till and app records, daily	the placeholder low, base and high range
Hourly demand	order times by hour, by day type	the hourly shape and peak the roster is built on
Conversion, where measurable	store-locator views or footfall against orders	the first link from score to cups
Service times	order to hand-over, per station	station seconds in the counter simulation
Labour needed	crew against waits and walk-aways each hour	the staffing rule and minimum crew
Machine cost and uptime	the lease quote, downtime log	the A$ machine cost and the setup choice

Then rerun: the area ranking with the area's real outcome, the setup comparison with real demand and times. Once several outlets trade, outcomes can train the supervised model; until then, weights stay a judgement.

EVIDENCE TO ACT · HELD 4 WEEKS IN A ROW

Example result, illustrative: 556 cups a day, 0.0% peak walk-aways, a quote of A$2,000.

- **Keep the pilot running.** 556 cups a day against about 236 (machine-made drink, University precinct).
- **No extra staff.** Peak walk-aways 0.0% against the 2% limit.
- **Keep machine-made drink.** Quote A$2,000 against the flip at about A$3,258 a month.
- **Open the next outlet, CBD office core.** It keeps 486 cups after 12.5% cannibalisation, against about 301 to break even.

EXPAND WHEN the pilot clears break-even for 4 weeks in a row, and the next area, the CBD office core, still clears it after losing about 13% of its cups to the pilot. Every figure here is a scenario assumption until the pilot replaces it.

Chagee's data collection: what is public, what is to confirm

Only what our sources say; everything else in the deck is synthetic or illustrative.

- **Public sources**
  - Chagee's annual report: *"Our business generates and processes personal, transactional and other data."*
  - The app: *"Order and pay digitally"*, *"Order for self-pickup ahead of time"*, *"Locate our outlets and their operating hours"*.
  - After the Singapore reopening queues, *"the brand rolled out in-app ordering and delivery"*.

  · Chagee Form 20-F, FY 2025 · CHAGEE App Store listing

  · Vulcan Post, Oct 2024

- **Chagee source, anonymous**
  - Singapore outlets run under **one company and one data system**.
  - All delivery is through **third-party apps**, so delivery orders arrive by platform.
  - People are deployed on **local expertise**: the GM and outlet teams make the staffing calls.

  An internal staff member who asked not to be named; paraphrased, not quoted.

- **To confirm**
  - The **app's share** of Singapore orders: not in the parts of the annual report we read.
  - Whether **searches or location** are logged. The app's privacy label lists precise location as not linked to the user and does not list search history (storefront seen: not Singapore).
  - **Wait times, footfall and walk-aways**.
  - Any use of app data needs **consent** under Singapore's data protection law (PDPA).

TAKEAWAY | **Orders are recorded in one system; searches, location and waits are to confirm. That is why our app and site data are synthetic (slide 10 and the appendix).**

From one customer's rule to the tipping point, and how far a friend network moves it

DERIVATION (OPIM639 TOPIC 8, SLIDES 30 TO 36)

Customer x in (0, 1) has interest r(x) = 1 - x. With a share z of friends participating, x joins if r(x) k z >= p, that is x <= 1 - p/(k z): the share who would join is 1 - p/(k z). An equilibrium is a share that reproduces itself, z = 1 - p/(k z), which gives **p = k z (1 - z)**. This hump is the equilibrium curve of the population; one customer's benefit is r k z, a straight line in z.

The hump peaks at k/4, at z = 50%. If p < k/4 the effort line crosses it twice: **z' = (1 - sqrt(1 - 4p/k))/2**, the unstable tipping point, and **z" = (1 + sqrt(1 - 4p/k))/2**, the stable level. At p = k/4 they merge at 50%; above it there is no positive equilibrium and participation falls to zero from any start. Rounds: z_next = max(0, min(1, 1 - p/(k z))).

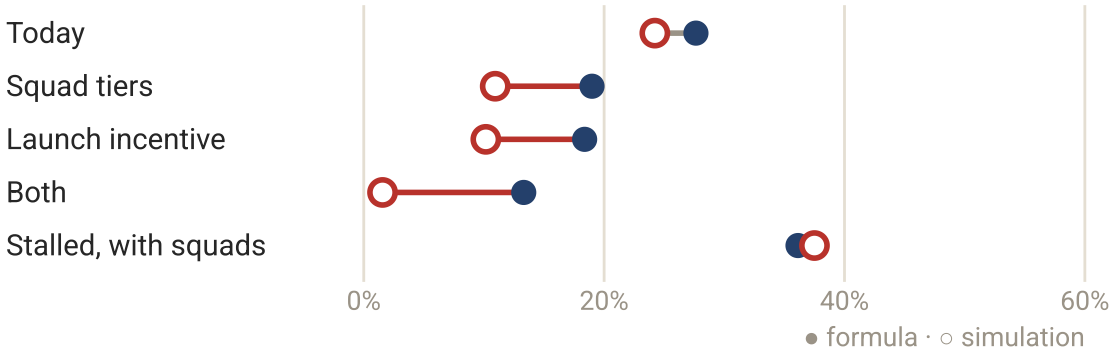
PARAMETERS (ILLUSTRATIVE INDEX, NOT CHAGEE DATA)

r individual interest, 0 to 1, uniform across customers. **k** value of friends participating; squads are hypothesised to raise it. **z** share of a customer's friends participating. **p** effort score of joining or progressing, 0 to 1: 1 is the whole benefit a fully interested customer gets with every friend on at k = 1; not money. Slider defaults: p 0.2, k 1, incentive 0, z0 0.35. Levers, read from the scenario table: squads add 0.3 to k; a launch incentive takes 0.05 off p.

SCENARIO TABLE, COMPUTED (SIM/MEMBERS.JS); THE BUILD FAILS IF IT DIFFERS FROM MEMBERS.EXPECTED_TABLE

Scenario	p	k	4p/k	z'	z"
Today	0.20	1.0	0.80	28%	72%
Squad tiers	0.20	1.3	0.62	19%	81%
Launch incentive	0.15	1.0	0.60	18%	82%
Both	0.15	1.3	0.46	13%	87%
Stalled app	0.30	1.0	1.20	none	none
Stalled, with squads	0.30	1.3	0.92	36%	64%

NETWORK CHECK · CLOSED FORM AGAINST THE FRIEND-NETWORK SIMULATION



Scenario	Closed form z'	Simulated	Gap, points
Today	28%	24%	-3.4
Squad tiers	19%	11%	-8.1
Launch incentive	18%	10%	-8.2
Both	13%	2%	-11.7
Stalled app	none	none	
Stalled, with squads	36%	38%	+1.4

2,000 customers on a seeded random graph with about 8 friends each, interest uniform; a customer is on next round if r k (share of friends on) >= p. 20 seeded runs; the simulated tipping point is the smallest starting share (to 0.01) from which the mean run ends above 10%. Gaps above five points mean the friend-group structure matters: with a handful of friends each, pockets of friends keep each other on, so the network tips lower than the formula, which assumes everyone sees the whole population's share. We report the gap, not hide it.

The break-even across the margin range, every new input, and what the sources say

BREAK-EVEN OF THE GROUP-ORDER CREDIT

Contribution per member a month = price x (margin x E - credit x (B + E)), with E extra orders and B group orders a member would place anyway. Break-even E* = credit x B / (margin - credit); none if the credit reaches the margin. Price S$5.90, credit 5%, B 2, E 1.

Gross margin	Extra orders needed a month	With AI cost (squad decision)	Contribution a member a month at E
50%	0.22	0.24	S$2.06
60%	0.18	0.20	S$2.65
70%	0.15	0.17	S$3.24

With personalised suggestions (squad decision) a flat AI cost a of S$0.05 a member a month is added: E* = (credit x B + a / price) / (margin - credit). Personalisation must add a / (price x (margin - credit)) = 0.02 orders a month on top of squads alone at the base margin.

NEW INPUTS (ASSUMPTIONS.JSON MEMBERS; ASSUMPTION, NOT CHAGEE DATA)

group credit share: 0.05 (range 0, 0.1). New input, assumption, not Chagee data.

extra orders per member month: 1 (range 0, 3). New input, assumption, not Chagee data.

base group orders per member month: 2 (range 0, 6). Invented for the break-even (the credit is paid on orders that would have happened anyway); assumption, not Chagee data.

ai cost per member month: 0.05 (range 0, 0.3). Illustrative placeholder, not sourced: the running cost of personalised suggestions per participating member, allocated. Replace with a quoted cost before any decision. Used only on slide 08 to show what personalisation must add on top of squads.

squad size: 2, 6. Our proposal, a design choice, not a Chagee feature claim.

example interest: 0.5. Display only: a middling customer, to separate one customer's benefit from the equilibrium curve.

app rules: points own drink 10, points per squad drink 5, points now 220, points next 300, tier now Silver, tier next Gold, people Mei, Jun, Aisha, Ravi, squad name Lunch squad, prefs Mei usual Jasmine milk tea · 50% sugar, suggest Oolong milk tea · 50% sugar, why you often pick floral teas at lunch and rated an oolong highly, Jun usual Black milk tea · less ice, suggest Black milk tea · 25% sugar, why you switched to less sugar in your last few orders, gift options Black milk tea · 25% sugar, Oolong milk tea · less ice, Fresh fruit tea, Aisha usual Fresh fruit tea, suggest Jasmine milk tea · no ice, why you liked jasmine teas when ordering for yourself, Ravi usual Oolong milk tea, suggest Oolong milk tea · less sugar, why close to your usual, with the sugar level you picked last week. Our illustrative rules for the proposed experience, used consistently on slide 06. Not Chagee's membership rules, which are to confirm. The group-order credit is members.group_credit_share. Preferences and suggestions are illustrative data, not learned from anyone. Recalling the last order is the baseline; the proposed AI part ranks alternatives from each member's own choices, acceptances and dismissals. No real messages, purchases or accounts.

network: customers 2000, friends per customer 8, runs 20, max rounds 200, z0 step 0.01, takeoff share 0.1. Assumption: eight friends each, a squad plus a few more. The closed form assumes everyone sees the whole population's share; a friend group sees a noisy sample of it.

SOURCE EXCERPTS · QUOTED WORD FOR WORD

> "we cannot accurately predict the rate or range of consumers who will join our membership program"
>
> Chagee Form 20-F, FY 2025, Item 3.D, risk factors

> "we may not be able to use our membership program to deepen our understanding of our target consumers"
>
> Chagee Form 20-F, FY 2025, Item 3.D, same risk factor

> "The value z' is the hump the firm must get over in order to succeed."
>
> OPIM639, Topic 8, Service Platforms, slide 34

> MEMBERSHIP OPERATIONS TEAM, CHAGEE
>
> Paraphrased, not quoted: the barrier to levelling up is not outlet operations; it is whether customers think the effort of tiering up is worth it.

How the enhanced forecast is built, and how it compares on synthetic windows

WHAT THIS IS

A synthetic demonstration of the mechanics only: one outlet's peak half-hour pickup window, with squad signals generated by our assumptions. The enhanced forecast can only help to the extent we assume the signals carry information, so this shows how the forecast would work, not measured accuracy. Nothing here feeds any other slide.

THE FORECASTS

Baseline: the calendar expectation for the window, cups a day x the peak hour's share x half an hour. **Enhanced**: the baseline's non-squad part, plus cups confirmed so far, plus predicted cups not yet confirmed: open baskets' unconfirmed invitees x the chance learned from synthetic history that they still confirm by then, unredeemed gifts x the learned chance of redemption, and the squad cups history says arrive from baskets and gifts not yet started. Each forecast uses only signals up to its time; a confirmed cup is counted once; an invitation or an unredeemed gift is never a confirmed drink.

MEAN ABSOLUTE ERROR, CUPS, OVER 400 SYNTHETIC WINDOWS

Squad share	120 min before: baseline / enhanced	60 min before: baseline / enhanced	30 min before: baseline / enhanced	10 min before: baseline / enhanced
0%	6.75 / 6.75	6.75 / 6.75	6.75 / 6.75	6.75 / 6.75
15%	6.86 / 6.83	6.86 / 6.28	6.86 / 6.16	6.86 / 5.95
30%	7.29 / 7.23	7.29 / 6.11	7.29 / 5.65	7.29 / 5.31
45%	7.79 / 7.70	7.79 / 6.13	7.79 / 5.19	7.79 / 4.63

With no squad share the two are identical. The enhanced forecast improves only because the synthetic squad signals were generated to move with demand; on single days it can be worse, when walk-in demand moves and squads cannot see it. Not measured accuracy.

SIGNALS AND WHERE THEY STAND

Documented: completed purchases (Chagee's Form 20-F: the business generates personal, transactional and other data; L-sec20f-data); in-app ordering at a chosen outlet (Vulcan Post, P-vulcanpost-comeback; App Store listing, P-appstore, not checked by us). **To confirm**: drink customisations and the pickup time chosen, in Chagee's Singapore app. **Proposed, needing collection and consent**: group size and confirmed quantities, suggestion accepted, edited or dismissed, gifts sent and redeemed, open baskets and invitations.

INPUTS (ASSUMPTIONS.JSON SQUADCAST; ASSUMPTION, SYNTHETIC, NOT CHAGEE DATA)

window minutes: 30 minutes in the pickup window. The window is the peak hour's first half (store.hourly_demand_share), so its calendar baseline is the deck's own volume: cups a day x the peak hour's share / 2.

lead minutes: 120 minutes before the window that the playback starts. Display and simulation horizon.

social share: 0.3 share of the window's cups that come through squad group orders and gifts, on average (range 0, 0.5). Hypothesis to test; zero means no squad signals, and the enhanced forecast then equals the baseline.

invites per basket: 4 people invited to a group basket, on average (squad size 2 to 6) (range 2, 6). Illustrative.

member conversion: 0.75 share of invited members who confirm a drink (range 0.3, 1). Illustrative; the forecast does not know it and estimates it from synthetic history.

confirm delay minutes: 8 average minutes from a basket opening to a member confirming. Illustrative.

gifts per window: 1.5 gifts sent for this window, on average, at the default squad share (scales with the squad share). Illustrative.

gift redeem share: 0.6 share of gifts redeemed for this window. Illustrative; an unredeemed gift is never counted as a drink.

gift redeem delay minutes: 30 average minutes from sending a gift to its redemption. Illustrative.

day shock sd: 0.2 spread of the day's demand around the calendar baseline (both walk-in and squad cups move with it). Illustrative: why the calendar baseline misses on any one day.

history windows: 400 past synthetic windows the enhanced forecast learns its conversion and late-arrival rates from. Synthetic history, same generator, different seeds.

eval windows: 400 synthetic windows used to compare the two forecasts (appendix). Different seeds from the history.

eval leads: 120, 60, 30, 10 minutes before the window at which the forecasts are compared. Appendix table.

day picks: 0.8, 1, 1.25 actual cups against the calendar baseline for the three example days on the slide (quiet, typical, busy). Display: the example days are the seeds whose actual cups, as a share of the baseline, are closest to these.

success pick: 1.45 actual cups against the calendar baseline for the fourth example day on slide 07 (squad rush). Display: a success case chosen by rule, not by hand. Among the same 300 synthetic windows, only those where the squad forecast is closer than the history-only forecast at the slide's default minute; of those, the one whose actual cups are closest to this share of the baseline. The slide also states the share of all windows where the squad forecast was closer, so the case is not presented as typical of every day.

drink mix: Jasmine milk tea 0.35, Oolong milk tea 0.3, Black milk tea 0.2, Fresh fruit tea 0.15 share of cups by drink, for predicted cups not yet chosen. Illustrative generic drinks, not Chagee's menu or sales mix.

Course link: OPIM639 Topic 1, slide 12 (capture demand data, not just sales data) and slide 13 (personalised service); Topic 2, slide 17 (the operator that provides fulfilment holds the detailed data). C-topic1, C-topic2.

Counter comparisons for every day type

FORECAST SLIDE · FORECAST ERROR: THE ROSTER FOLLOWS A WRONG FORECAST, THE CUSTOMERS DO NOT (TODAY'S SPLIT, BUCKET BRIGADE)

Forecast	Normal day crew · away · S$/day	Busy day crew · away · S$/day	Event day crew · away · S$/day
30% low	1 · 0.9% · S$372	1 · 20.8% · S$372	2 · 1.2% · S$402
15% low	1 · 0.9% · S$372	1 · 20.8% · S$372	2 · 1.2% · S$402
exact	1 · 0.9% · S$372	2 · 0.0% · S$402	2 · 1.2% · S$402
15% high	1 · 0.9% · S$372	2 · 0.0% · S$402	3 · 0.0% · S$417
30% high	1 · 0.9% · S$372	2 · 0.0% · S$402	3 · 0.0% · S$432

BUCKET BRIGADE (APPENDIX) · SHARING RULES AT THE FORECAST CREW, TODAY'S SPLIT

Rule	Normal day crew · away · wait	Busy day crew · away · wait	Event day crew · away · wait
Bucket brigade	1 · 0.9% · 2:08	2 · 0.0% · 1:14	2 · 1.2% · 2:38
Fixed posts, fastest on toppings	1 · 1.0% · 2:10	2 · 0.0% · 1:14	2 · 4.1% · 4:10
Fixed posts, slowest on toppings	1 · 1.0% · 2:10	2 · 0.7% · 3:26	2 · 34.9% · 9:47

AUTOMATION (APPENDIX) · FIVE SETUPS: THE FEWEST PEOPLE THAT KEEP WALK-AWAYS WITHIN 2%

Setup	Normal day people · away · wait · S$/day	Busy day people · away · wait · S$/day	Event day people · away · wait · S$/day
Today's split (our scenario)	1 · 0.9% · 2:08 · S$372	2 · 0.0% · 1:14 · S$402	2 · 1.2% · 2:38 · S$402
Machine-made drink (scenario)	1 · 0.0% · 1:14 · S$435	1 · 0.0% · 1:28 · S$435	2 · 0.0% · 1:06 · S$465
All by hand	2 · 0.3% · 1:16 · S$361	3 · 0.0% · 1:13 · S$392	4 · 0.0% · 1:24 · S$422
All robots	– · 0.0% · 0:52 · S$562	– · 0.0% · 0:52 · S$562	– · 0.0% · 0:58 · S$562
One machine (tea and milk)	2 · 0.0% · 1:10 · S$305	2 · 0.1% · 1:36 · S$305	3 · 0.2% · 1:44 · S$335

THE FORECAST'S EFFECT AT THE COUNTER: NORMAL-DAY ROSTER → FORECAST ROSTER, SAME EQUIPMENT AND CUSTOMERS

Setup, bucket brigade	Normal day crew · away · S$/day	Busy day crew · away · S$/day	Event day crew · away · S$/day
Today's split (our scenario)	1→1 · 0.9%→0.9% · S$372→S$372	1→2 · 20.8%→0.0% · S$372→S$402	1→2 · 53.8%→1.2% · S$372→S$402
Machine-made drink (scenario)	1→1 · 0.0%→0.0% · S$435→S$435	1→1 · 0.0%→0.0% · S$435→S$435	1→1 · 5.4%→5.4% · S$435→S$435
All by hand	2→2 · 0.3%→0.3% · S$361→S$361	2→3 · 5.8%→0.0% · S$361→S$392	2→4 · 40.1%→0.0% · S$361→S$422

Peak hour, mean of the seeded runs; cost is the day's roster and machines. Illustrative assumptions, not Chagee data.

 TBC · DRAFT, TO BE CONFIRMED

Four modelled decisions by owner: where to open for the expansion manager; what to automate, how many people and how to share for the operations manager

	DECISION	WHY, FROM THE MODEL	REVISIT WHEN
1 **Where to open** expansion manager · slide 10 · DBSCAN + assumed demand model	**Open where app searches cluster, not beside an existing outlet.**	**+731** against **+216** cups a day for the same new outlet. Synthetic app-search data: this shows the method, not a finding about real outlets.	Rerun on Chagee's app logs, if searches and location are recorded, before any lease.
2 **What to automate** operations manager · appendix · physical AI	**In our model, one tea-and-milk machine is the cheapest on every day type; pilot it against today's split (our scenario).**	**S$305–335** a day against S$372–402. The two cost the same at about **S$905 a month per machine unit** (1 unit against 3), with wage, volume, station speeds and crews held fixed; we assume S$1,909. Amacha's S$10,000 is a one-off price; without machine life, upkeep and financing we cannot turn it into a monthly cost.	When Chagee's real machine cost is known.
3 **How many people** operations manager · forecast slide and slide 13 · forecast + economic rule	**Economic rule: add a person on event days, not on busy days.**	With the one machine: peak crew 2 → **3** on event days; 2 on busy days, where an extra person costs more than the orders it keeps.	When the forecast peak moves.
4 **How to share** operations manager · appendix · operations	**Test a bucket brigade against fixed posts for the hand steps.**	In the recommended setup (one machine, 3 people, event day): **0.2%** walk away with a brigade, 0.4% with well-assigned fixed posts, 22% with badly assigned ones; median wait 1:44 against 2:18.	Fixed posts once there is spare staff.

- **Pilot in 3 outlets**

Shadow	Roster	Machine
forecast only	economic rule	one vs today's split

KPIs: cups per labour hour, wait time and walk-aways at the peak. Compare forecast and actual cups weekly.

- **Limits we know**

All figures are illustrative assumptions, not Chagee data. Machine costs and speeds are assumed; breakdowns and breaks are not modelled.

- **To confirm with Chagee**

The real machine cost per unit, whether outlets log wait times and walk-aways, and how today's local staffing calls compare with the forecast.

ASSUMED BASELINE illustrative — One machine and a flat crew of 2 every day: **23%** of event-day peak orders walk away.

PROPOSED SCENARIO in our model — The forecast and the economic rule add a person on event days only (2 → 3): **0.2%** walk away; busy days stay at 2 (0.1%).

Per outlet, about 61 orders kept on an event day and 5 on a busy day; on an illustrative calendar, S$7,539 of revenue a year

Modelled avoided costs on stated assumptions, not demonstrated savings, against an illustrative baseline, not Chagee's operations. Economic rule for the headline figures. Peak hour only; conservative. Prices to our knowledge, based on public sources.

Profit a year, 44 outlets: **S$1.09m** = counter automation, physical AI **S$1.04m** (depends on the minimum-crew rule) + forecast roster, predictive AI **S$0.05m** | Without / with the minimum-crew rule: **S$0.37m to S$1.09m**

ONE OUTLET AT 556 CUPS A DAY · BOTH LEVERS, ONE RUN

EVENT DAY
61 orders kept
Revenue S$358 (× S$5.90) · profit S$241
Counter: labour saved − machine S$27

BUSY DAY
5.4 orders kept
Revenue S$32 (× S$5.90) · profit S$76
Counter: labour saved − machine S$57

NORMAL DAY
0.2 orders kept
Revenue S$1 (× S$5.90) · profit S$58
Counter: labour saved − machine S$57

A YEAR	PER OUTLET	44 OUTLETS, IF ALL LOOKED LIKE THIS MODEL
Orders kept peak hour, busy and event days	**1,278** range −121 to 1,452	**56,221**
Revenue orders × price	**S$7,539** range −S$832 to S$10,015	**S$331,704**
Profit margin 50%–70%: −S$754 to +S$754	S$24,660	S$1,085,027
Labour saving, counter labour saved − machine cost; shown on its own	S$20,136	S$886,005

Revenue and labour are never added: revenue is orders × price; profit is orders × price × margin, plus labour saved, minus machine cost.

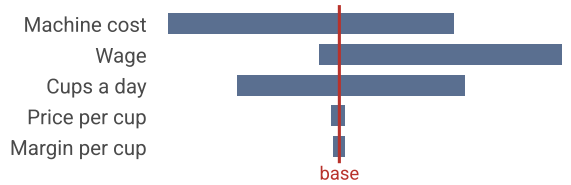
Calendar: 104 busy days (weekend days) and 11 event days (Singapore public holidays) a year: an illustrative definition, to confirm with Chagee; sliders on screen.

ORDERS · PROFIT A YEAR	WALK-AWAY RULE	ECONOMIC RULE
Forecast crew	1,242 · S$603	670 · S$1,711
Machine	888 · S$23,611	888 · S$23,611
Both, one run	**1,278 · S$24,660**	**1,278 · S$24,660**
Both, today's split counter −S$305/month	1,124 · −S$3,142	1,124 · −S$3,142

Walk-away rule: fewest people keeping peak-hour walk-aways within 2% (the forecast crew at 556 cups a day). Economic rule: add a person only when orders kept × price × margin beat the extra labour. "Both" is one run, not a sum: the levers keep many of the same orders, and with the machine the forecast crew needs fewer extra hours (−S$662 profit against the sum). Headline figures use the economic rule.

Against what
A flat normal-day roster and an all-by-hand counter. Chagee deploys people on local expertise (Chagee source, anonymous), so this baseline is ours, not Chagee's.

All inputs at their worst/best end together
Profit a year, one outlet: low **−S$15,660** · base **S$24,660** · high **S$101,166**



Fragile: the minimum crew
The counter lever rests on the model's minimum crew off-peak: 2 by hand, 1 with machines. Profit a year per outlet: S$24,660 with it; S$8,395 with 1 either way.

Not sized
Delivery levers (third-party; channel margins not disclosed), membership and CRM, waste (no inventory model in this deck), and the new-store site forecast (shown as cups added only).

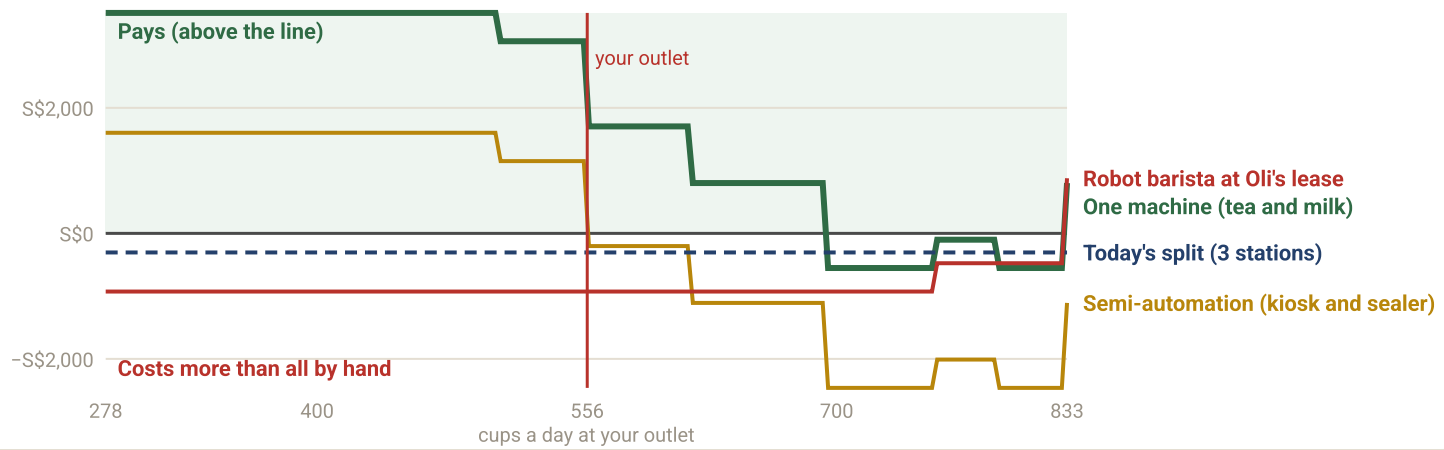
ASSUMED BASELINE illustrative — Flat normal-day roster, all by hand: about **1,291** peak-hour orders a year walk away per outlet.

PROPOSED SCENARIO in our model — Both levers, one run: **1,278** kept a year per outlet, profit **S$1.09m** across 44 outlets (S$0.37m without the minimum-crew rule).

A machine pays only when it removes a person-shift for less than the shift costs

To our knowledge, based on public sources; vendor list prices, not cost to implement. Chagee's equipment cost is not public; its Singapore relaunch used semi-automated equipment (Vulcan Post).

NET GAIN A MONTH AGAINST ALL BY HAND · ONE OUTLET



Shown at 556 cups a day, wage S$15.06 an hour, S$1,909 a month per machine unit (sliders on screen).

1 Semi-automation	2 Beverage platform	3 Milk-tea machine	4 Robot arm	5 Robot barista
Gong Cha Singapore	Gong cha Super Wu	Amacha, one shop	Richtech ADAM	Oli; coffee-first
Not disclosed	*Not disclosed*	S$10,000 to buy	*Not disclosed, rental only*	US$4,990/mo lease
PRESS	COMPANY CLAIM	PRESS	VENDOR PAGE	VENDOR PAGE

The model's S$1,909 a month is per station unit (our reading: a semi-automated station). Semi-automation and the beverage platform are priced at it because their real prices are not disclosed; robot prices are per setup. Full table and sources in the appendix.

START HERE

One machine on the tea and milk

+S$1,706 a month vs all by hand

Saves 8 person-hours a day (S$3,614 a month) for one S$1,909 unit. Simple payback, if it lets the roster drop one person-shift: 2.8 months on Amacha's S$10,000.

WATCH

Today's 3-station split

−S$305 a month vs all by hand

Saves 12 person-hours a day (S$5,422) but pays for 3 units (S$5,726).

WAIT FOR VOLUME

Robots: arm or barista

757+ cups a day before they save more labour than today's split (labour only, not a cost break-even)

At 556 cups, Oli's lease (S$6,350) is S$928 a month worse; a robot setup would need to cost under S$5,422 a month. The arm's price is not disclosed.

Model assumption that drives all three: minimum crew 2 by hand, 1 with machines, off-peak. Station speeds are the model's, not vendor claims. Rosters step in whole people, so results flip with volume and wage.

WITHOUT AI
our assumed baseline

All by hand at 556 cups a day: **24 person-hours** a day, **S$10,843** a month of labour.

WITH AI
our proposal

One tea-and-milk machine: **16 person-hours** a day, **+S$1,706** a month after its unit cost. A robot setup (four stations on machines) starts to save more labour than today's split at about 757 cups a day; labour only, not a cost break-even.

Every rung, every price as published, and where it comes from

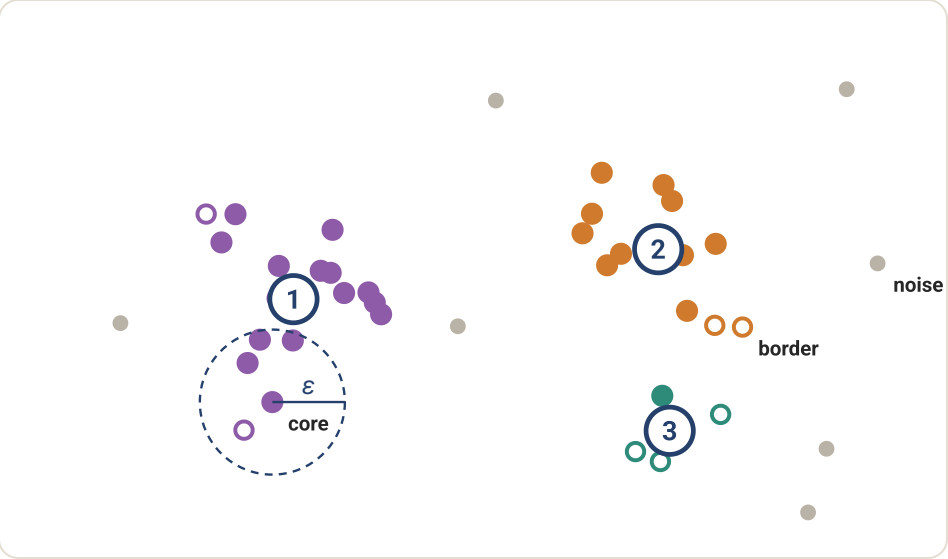
To our knowledge, based on public sources. Physical AI: automation and machines. A gap is "not disclosed", never a guess.

RUNG	EXAMPLE	AUTOMATES · STEPS ON MACHINES (OUR MAPPING)	VENDOR LIST PRICE	CLAIMS (NOT MODEL INPUTS)	RELIABILITY	SOURCE
1 **Semi-automation**	Gong Cha Singapore	Sugar dispensers, cup sealers, self-order kiosks. **Order and tag, Shake and seal**. Kiosk = order and tag, sealer = shake and seal. Sugar dispensing sits inside tea and milk, so it is not modelled.	*Not disclosed*	Over 20% productivity from self-ordering; 5% manpower saving (no baseline given)	PRESS	https://www.ntuc.org.sg/uportal/news/Technology-Brewing-Up-the-Bubble-Tea-Business/
2 **Beverage platform**	Gong cha Super Wu (Feb 2026)	Dispenses tea, sweeteners and syrups; brewing and shaking stay manual. **Tea and milk**	*Not disclosed*	Up to 65% higher peak productivity	COMPANY CLAIM	https://marketech-apac.com/gong-cha-launches-gong-cha-2-0-to-modernise-global-operations-and-store-design/
3 **Single automated milk-tea machine**	Amacha (one small shop, owner interview)	Makes the milk tea itself. **Tea and milk**	S$10,000 to buy	The shop's total opening cost was S$209,000 (not a machine cost)	PRESS	https://www.malaymail.com/news/life/2026/10/05/opening-a-milk-tea-shop-in-singapore-can-rack-up-to-s209000-even-before-the-first-cup-is-poured/237715
–	**Our scenario: today's split**	**Order and tag, Tea and milk, Shake and seal**. A scenario of ours (the appendix counter slides), not Chagee's disclosed setup	Model: S$1,909 a month per station unit		OUR ASSUMPTION	assumptions.json
4 **Robot arm that makes boba**	Richtech ADAM	Cup, ice, tea, toppings and sealing by one robot arm. **Cup and ice, Tea and milk, Toppings, Shake and seal**	*Not disclosed*. Rental only; price not published	–	VENDOR PAGE	https://richtechrobotics.com/solutions/adam
5 **Robot barista**	Oli	A whole drink station in one unit. **Cup and ice, Tea and milk, Toppings, Shake and seal**. Coffee-first. No tea drinks listed for Oli or Artly.	US$59,000 (≈ S$75,078) to buy; US$4,990 (≈ S$6,350) a month lease	–	VENDOR PAGE	https://olirobotics.com/best-robot-barista
	Artly		US$68,000–88,000 (≈ S$86,530–111,980) to buy; US$2,500–4,000 (≈ S$3,181–5,090) a month lease; US$2,500–4,000 a month software Oli's estimate of a competitor. The figures as given do not say whether the lease replaces or adds to the purchase, so no verdict is computed.	–	VENDOR PAGE	https://olirobotics.com/best-robot-barista
	Cafe X		US$250,000 (≈ S$318,125) to buy Oli's estimate of a competitor's price.	–	VENDOR PAGE	https://olirobotics.com/best-robot-barista
–	**Chagee Singapore**	Chagee's own equipment cost is not public. Its Singapore relaunch used semi-automated equipment.	*Not public*	–	PRESS	https://vulcanpost.com/872176/chagee-singapore-comeback-after-sudden-exit/

Vendor list prices, not cost to implement: installation, training, maintenance, power, space and import are not included; USD converts at S$1.2725 per US$. **Simple payback** = list price ÷ labour saved a month, with no maintenance or financing. **The model's S$1,909 a month** is per station unit: our reading: one semi-automated station unit (for example a dispenser or a cup sealer), amortised with upkeep. Not a robot barista lease; robot prices in machine_tiers.json are per setup. **Vendor speed and throughput claims** are listed, never used: every rung runs at the model's station speeds. **Steps on machines** is our mapping. Artly and Cafe X prices are Oli's estimates of competitors.

DBSCAN finds where searches bunch up; a proximity model turns a pin into cups

DBSCAN IN FOUR MOVES · TOY EXAMPLE



1. **Radius ε.** Two searches are neighbours if they are within ε of each other (the cluster-radius slider).
2. **Core.** A search with at least the minimum number of searches within ε (counting itself) is a core point.
3. **Grow.** A cluster spreads from core to core. Searches it reaches that are not core join as border points.
4. **Noise.** Searches no cluster reaches. Unlike k-means, nobody picks the number of clusters, and odd shapes are fine.

Unsupervised: no labels or past openings are needed. Ester, Kriegel, Sander and Xu (1996). On slide 10 it runs only on searches with no outlet within walking distance; each cluster's centre is a candidate site.

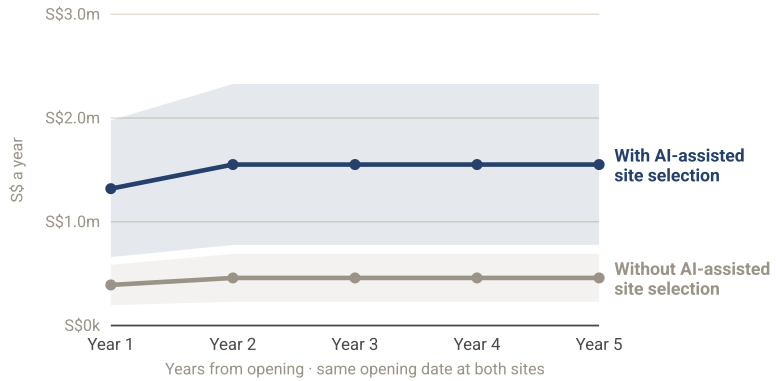
THE SIMULATED NUMBERS, IN ORDER · AT THE SLIDER DEFAULTS

1	**App searches**	Synthetic dots on land: 15% spread evenly, 35% around the outlets' malls, the rest around 20 towns	**3,000 dots**
2	**No store nearby**	The nearest outlet is further than the walking distance	**1 km · 46% of dots**
3	**DBSCAN**	Clusters the no-store-nearby dots	**ε 0.6 km · min 15 · 11 clusters**
4	**Pull of a store**	Fades with distance d: $\exp(-½\,(d/L)^2)$, with the length L set by the walking distance	**L = 1 km**
5	**Who buys where**	A dot buys at a store in proportion to its pull, against the pull of not buying (a store next door = one)	**not buying 0.3**
6	**Calibration**	Each dot's weight is set so the average outlet sells the deck's cups a day (cups a year ÷ open days)	**556 cups a day**
7	**Score a pin**	Cups for Chagee = what the new store sells − what it moves from nearby outlets	**top: 837 − 106 = +731**
8	**Verdict**	Share of the new store's cups that are new to Chagee	**strong ≥ 60% · weak < 40%**
9	**Opening roster**	Peak-hour cups ÷ cups one person handles, with the safety margin	**one per 83 cups**

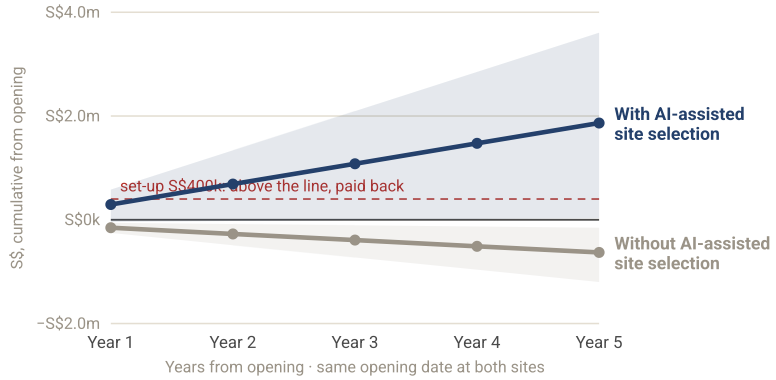
All synthetic: the dots, towns and weights are assumptions on the team's map (not to scale), not Chagee app logs or population data. With real app logs, steps one and two change; the rest runs unchanged.

Same new outlet, same opening, same counter: the AI site adds about S$2.5m more operating profit over 5 years, in our model

ANNUAL INCREMENTAL REVENUE TO CHAGEE, S$



CUMULATIVE INCREMENTAL OPERATING PROFIT, S$



THE BRIDGE, FIVE-YEAR TOTALS, BASE SCENARIO

5 YEARS, BASE SCENARIO	WITHOUT AI · +216	WITH AI · +731	DIFFERENCE
New-store sales	S$6.2m	S$8.6m	S$2.4m
− Sales displaced from nearby Chagee outlets	−S$4.0m	−S$1.1m	S$2.9m
= Net additional network sales (incremental revenue)	**S$2.2m**	**S$7.5m**	**S$5.3m**
− New-store ingredients, packaging and delivery commission	−S$3.1m	−S$4.2m	−S$1.2m
− New-store labour (same staffing rule)	−S$325k	−S$412k	−S$87k
− Machines, rent, utilities, maintenance	−S$1.4m	−S$1.4m	S$0k
+ Nearby outlets' variable costs saved (their rent and staff stay)	S$2.0m	S$537k	−S$1.4m
− AI recurring cost (allocated)	S$0k	−S$120k	−S$120k
= Incremental operating profit, before depreciation, interest and tax	**−S$630k**	**S$1.9m**	**S$2.5m**

Illustrative scenarios on our assumptions, not a demonstrated causal gain from AI. DBSCAN finds the search clusters; the assumed demand model estimates the sales. Shaded: the scenario range (our per-outlet volume range), not a confidence interval.

5-YEAR RESULTS · BASE · ILLUSTRATIVE

Revenue uplift	S$5.3m
Operating-profit uplift	S$2.5m
Payback, with AI	1.4 years
Payback, without AI	beyond year 5

Cumulative cash flow after the S$400k set-up (AI site S$460k). AI cost repaid in 2 months. Volume range: S$1.2m to S$3.8m profit uplift.

ASSUMPTIONS · S$ · REUSED UNLESS MARKED NEW

360 days · S$5.90 a cup, held flat · ingredients and packaging 40%
delivery 30% of cups at 30% · wage S$15.06 an hour, same staffing rule
rent S$15,000, utilities S$2,500, maintenance S$800 new a month · same machines
year-one ramp-up 85% new · growth 0.0% new · cost inflation 0.0% new
AI S$60k to build, S$24k a year, 100% on this outlet new · set-up S$400k, outside operating costs

TAKEAWAY

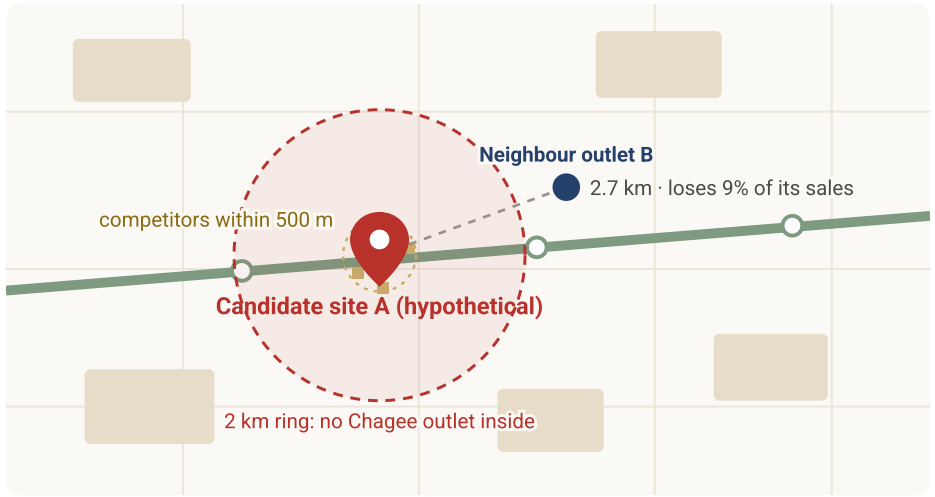
Beside the nearest outlet: **+216** net cups a day → about **S$2.2m** of network sales over 5 years, and **−S$630k** operating profit: the contribution lost at its neighbours outweighs what it makes. With AI-assisted site selection: **+731** net cups a day → about **S$7.5m** and **S$1.9m** after the AI costs: **S$2.5m** more profit than the baseline. Illustrative, not a proven causal gain.

Extension: where this scales next. Not included in the S$ benefit.

Four scores for every location

Corporate drops a pin; the model rates it on four things (illustrative). Decision owner here: the expansion manager.

SCHEMATIC MAP · HYPOTHETICAL PINS



Demand score HAND-SET
Residents, workers and transit riders within reach, plus footfall. — 84

Delivery demand score HAND-SET
Orders from the area on third-party delivery apps.
All Chagee delivery goes through third-party apps (anonymous Chagee source, to confirm), so we use area demand, not "served from a farther outlet". Whether the platforms share this data is to confirm. — 78

Competition score HAND-SET
Bubble-tea and cafe outlets within 500 m; dense competition lowers the score, a few prove demand. — 61

Cannibalisation factor COMPUTED · SLIDE 10 ENGINE
Site A: 13% of its cups would move from existing Chagee outlets; neighbour outlet B loses 9%. Deducted so the result is net to Chagee. — −13

Site score WEIGHTED
Demand 50%, delivery 20%, competition 30% (hand-set weights). — 76

ATTRIBUTES THE MODEL TAGS TO EACH PIN residents and workers within 1 km · MRT and bus tap-ins · footfall · delivery orders from the area · competing outlets within 500 m · Chagee outlets within 2 km · rent per sq ft · staff needed at peak

WHAT IS AI HERE Scoring a pin from its attributes would be a supervised model trained on outlets' results: a production idea. Here the three scores are set by hand, and the cannibalisation factor is computed by the slide 10 engine on synthetic data. The next slide runs the supervised model.

Illustrative. Synthetic data, not Chagee data. Chagee's opening history is internal and we do not have it.

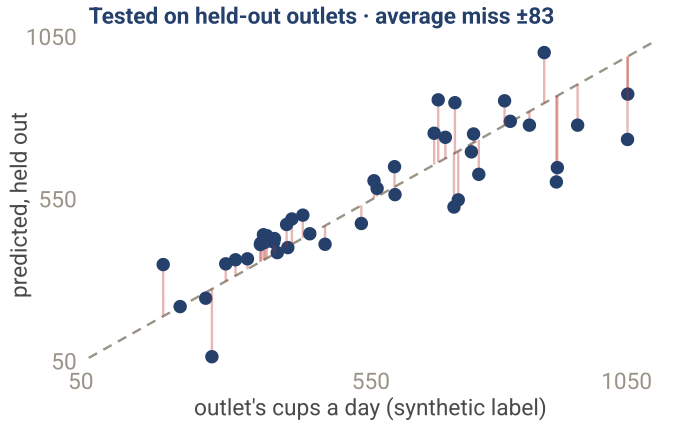
Extension: where this scales next. Not included in the S$ benefit.

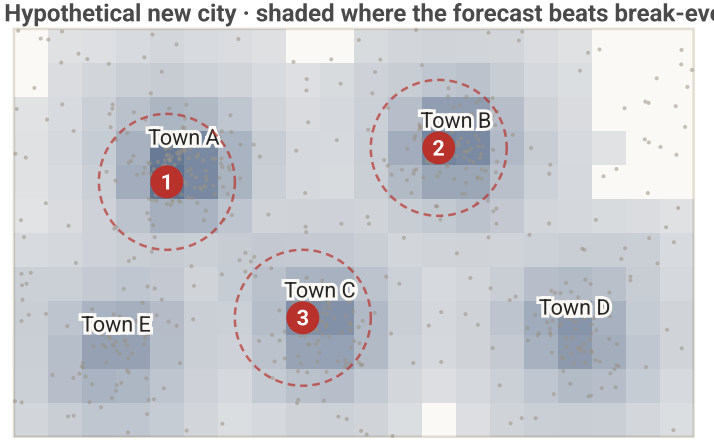
Trained on Singapore's outlets, the model picks where to open in a city where Chagee has no data yet

Supervised learning: keep only the inputs a new city has, test the model on outlets it has not seen, then let it score every spot in a hypothetical city (illustrative, synthetic).

TRAIN ON 44 OUTLETS · TOGGLE THE INPUTS

- ☑ **People within reach** — NEW MARKET: HAS IT
- ☑ **Distance to a town centre** — NEW MARKET: HAS IT
- ☐ **Delivery-area demand** — NEW MARKET: TO CONFIRM
- ☐ **Unserved app searches** — NEW MARKET: NONE
- ☐ **Chagee outlets nearby** — NEW MARKET: NONE
- ☐ **Distance to nearest Chagee outlet** — NEW MARKET: NONE
- ☐ **Competitors within 500 m** — NOT IN OUR DATA
- ☐ **Rent per sq ft** — NOT IN OUR DATA





TESTED ON OUTLETS IT HAS NOT SEEN

±83 cups a day, average miss

R^2 0.76 with 2 inputs

All inputs (Singapore)	**±41 · R^2 0.95**
New-market inputs only	**±83 · R^2 0.76**
No inputs: guess the average outlet	**±197**

WHERE TO OPEN IN THE NEW CITY

1. Town A: ≈ **1,090** ± 83 · **above break-even even at 1,007** · **beyond what it learned**
2. Town B: ≈ **959** ± 83 · **above break-even even at 876**
3. Town C: ≈ **887** ± 83 · **above break-even even at 804**

Cups a day, forecast ± average miss. Break-even ≈ 275 at Singapore costs (the pin-to-profit slide), a stand-in. Picks ≥ 2 km apart.

WHAT IS AI HERE Supervised learning (ridge regression) from site attributes to cups a day. The labels are the slide 10 engine's synthetic outlets, so the fit flatters; Chagee's real opening history would fit worse. Using Singapore to score another market assumes its customers behave alike: check against the first outlets' actuals.

Illustrative. Synthetic data, not Chagee data. Chagee's opening history is internal and we do not have it.

Extension: where this scales next. Not included in the S$ benefit.

From pin to net profit

Six steps from a pin to a go / no-go, on profit net to Chagee overall (illustrative). Decision owner here: the expansion manager.

1 Pin a location
Corporate clicks a point on the map.
NOT AI

2 Tag attributes
People, transit, footfall, delivery, competitors, outlets nearby, rent.
DATA

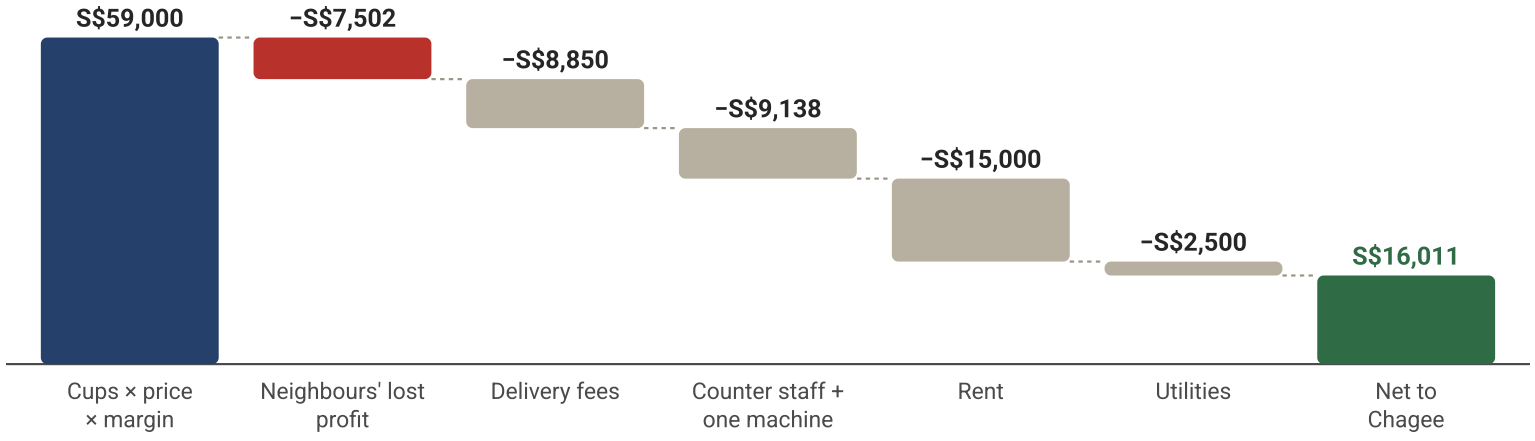
3 Four scores
Demand, delivery, competition, cannibalisation (previous slide).
AI · SUPERVISED MODEL

4 Find look-alikes
Anonymised outlets (Outlet 12, Outlet 27): their cups a day, what neighbours lost.
AI · SIMILARITY SEARCH

5 Forecast net profit
Cups a day, running cost, minus the profit neighbours lose.
COMPUTED BELOW

6 Decide
Open, pilot or skip. Actuals feed back: continual learning.
AI · CONTINUAL LEARNING

NET PROFIT TO CHAGEE, A MONTH · SITE A, TOP APP-SEARCH CLUSTER



At 278–833 cups a day: **−S$3,506** to **S$34,624** a month. Neighbour outlet B, 2.7 km away, loses 9% of its sales. Delivery fees: 30% of cups × 30% commission. Counter: the one-machine setup, normal-day roster (counter model).

STEP 6 · RECOMMENDATION

Pilot

positive at 556 cups a day, negative at 278, the low end: test it before a full lease.

Site score **76 / 100 (four scores)**
Payback **≈ 25 months on S$400,000**

A strong-demand site can still fail if rent or staffing is too high, or if it mostly takes sales from our own neighbours.

In practice this needs Chagee's opening history. With about 44 outlets in Singapore the model must stay simple (few attributes, ranges not point values).

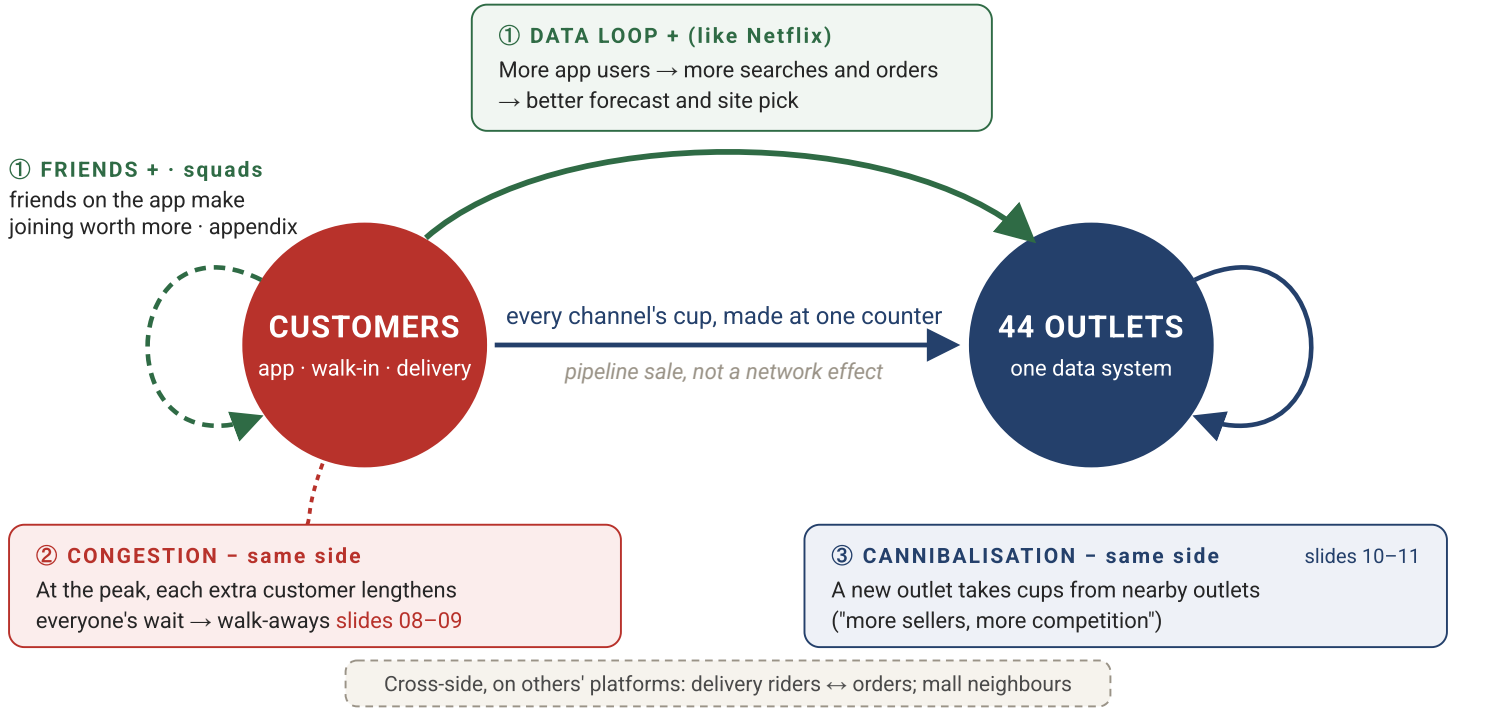
Illustrative. Synthetic data, not Chagee data. Chagee's opening history is internal and we do not have it. Cups a day, price and margin are the benefit slide's; rent, delivery, utilities and set-up cost are assumptions, not Chagee data.

Course link · cross-side vs same-side network effects (eBay)

Chagee's network effects sit on the customer side: we grow the friend and data loops, then cap the crowding they bring

Our reading: Chagee sells its own tea, so there is no seller side to match. Part one grows the positive loops; parts two and three cap the crowding.

WHERE CHAGEE SITS ON THE DIAGRAM · CLICK A LOOP



① GROW · PART ONE · SQUADS, APPENDIX

Make friends worth more, feed the data loop

Squads: friends taking part raise the features' value (not AI). **Why Chagee:** app-first and member-led: **257 million** members group-wide; repeat members place **more than 78%** of orders (earnings call, Aug 2026; SG to confirm).

② CAP · PART TWO · SLIDES 08 AND 09

Absorb the peak crowd

Forecast crew plus a bucket brigade: event day walk-aways **54%** → **1.2%** of peak-hour orders.

③ CAP · PART THREE · SLIDES 10–11

Count cups added, not sold

Of 837 cups at the new site, **106** (13%) move from nearby outlets. The score subtracts them.

NOT OUR LEVER

Delivery apps and malls own the cross-side effects; their riders still count in our peak.

Takeaway | Grow the friend and data loops, then make sure the counter and the outlet map can absorb the extra customers.

Where every number comes from

FACTS QUOTED

Outlets in Singapore: 44. Store count used on the storyboard deck. The doc also cites 46 (VnExpress, earlier); the later directory count is used. *[L-sg-directory]*

Global context, group-wide, not Singapore: franchise share of FY2025 net revenue: 0.88. Secondary source; flagged wherever shown. GROUP-WIDE: global context only, never applied to Singapore (Singapore is company-owned, I-sg-owned). *[L-koalagains]*

Global context, group-wide, not Singapore: company-owned share of FY2025 net revenue: 0.12. Secondary source; flagged wherever shown. GROUP-WIDE: global context only, never applied to Singapore (Singapore is company-owned, I-sg-owned). *[L-koalagains]*

Registered members, group-wide, end of June 2026: 257000000. Company statement on the Q2 2026 earnings call (COO, interpreted). Group-wide, not Singapore. Membership is on Chagee's mobile mini program (L-chagee-q1-2025). *[L-chagee-q2-2026-call]*

Orders from members with two or more purchases, group-wide (more than): 0.78. Company statement on the Q2 2026 earnings call. A floor ('more than 78%'). Group-wide, not Singapore. *[L-chagee-q2-2026-call]*

SOURCES

L-sg-directory Chagee Singapore store directory, mapped by the team (44 locations, checked 15 Oct 2026); Mothership, Sep 2026, https://mothership.sg/2026/09/chagee-more-heartland-stores-singapore-eugene-lee/ gives the same count. Taken from the team's ideation doc (OPIM 639_ Ideation Process), not re-checked for this deck.

I-sg-owned PROJECT OWNER CONFIRMATION (23 Oct 2026): Chagee Singapore operates a company-owned model, not a franchise model. Confirmed by the project owner; not from the anonymous Chagee contact and not an external citation. Used on slide 01 (headline, step 1, card, notes). Supersedes the earlier 'ownership not claimed' framing. It does not change the group-wide revenue shares, which stay global context only.

L-koalagains KoalaGains, 'CHA business and moat', https://koalagains.com/stocks/NASDAQ/CHA/business-and-moat (SECONDARY - flag when used): FY2025 net revenue about 88% from the franchise (master franchise) model and about 12% from company-owned stores, the latter growing fastest. Taken from the team's ideation doc (OPIM 639_ Ideation Process), not re-checked for this deck.

L-yfchuhai YF Chuhai, https://www.yfchuhai.com/article/9751.html: Chagee's leaf supply chain is in China; it sources only two ingredients locally, fresh milk and fruit. Taken from the team's ideation doc (OPIM 639_ Ideation Process), not re-checked for this deck.

L-sec20f Chagee Holdings Form 20-F for FY2025, SEC EDGAR, https://www.sec.gov/Archives/edgar/data/2013649/000110465926050766/cha-20251231x20f.htm: separate centralized and regional warehouses to lower storage and logistics costs. Taken from the team's ideation doc (OPIM 639_ Ideation Process), not re-checked for this deck.

L-unsw UNSW BusinessThink, Oct 2024: Chagee semi-automation, baristas scan a QR code and the machine dispenses the formulation. (brief.md section 9)

I-chagee-contact CHAGEE SOURCE, ANONYMOUS (Oct 2026): a Chagee source who asked not to be named; do not identify them. Four points only: whether Singapore outlets are owned, a joint venture or franchised does not matter for this work, because the data sits in one system; Singapore outlets run under one company and one data system; all delivery is via third-party apps; people are deployed on local expertise: the GM and outlet teams make the staffing calls. Ownership itself is not attributed to this source: see I-sg-owned (project owner confirmation).

I-sg-groundfloor INFORMAL - quote in the team's ideation doc (Chagee Singapore's Lee: 'Whenever you see a Chagee, it's always on the first floor'; kiosk-style outlets in basements and MRT stations at entry in 2019). The article is not named in the doc; confirm the source before quoting.

I-team-app-first INFORMAL - team observation in the ideation doc: Chagee Singapore sells mainly through its app, with minimal order kiosks in outlets. Confirm before citing. SUPERSEDED on slide 01 (22 Oct 2026) by P-appstore, P-vulcanpost-comeback and L-sec20f-data; the app's share of orders stays to confirm.

L-ester1996 Ester, M., Kriegel, H.-P., Sander, J. and Xu, X. (1996). A density-based algorithm for discovering clusters in large spatial databases with noise. Proceedings of the Second International Conference on Knowledge Discovery and Data Mining (KDD-96), 226-231. The DBSCAN algorithm used on slide 13.

P-honeycombers PRESS - in-store sightings, indicative. Honeycombers (Feb 2026), https://thehoneycombers.com/singapore/china-bubble-tea-brand-singapore/ : Chagee price sightings in Singapore; not a current full menu.

P-eatbook PRESS - in-store sightings, indicative. Eatbook (Aug 2024), https://eatbook.sg/?p=168476 : Chagee price sightings in Singapore; not a current full menu, and older than the deck.

I-team-expansion INFORMAL - team research (Oct 2026): Chagee is looking to expand rapidly in Singapore; how many stores it is going for is confidential. Singapore only. Confirm before citing.

I-team-site-framework TEAM WORK (Oct 2026): Jin's site-selection framework, two team slides ('Four Scores for Every Location', 'From Pin to Profit Forecast'): four scores per pin, six steps from a pin to a go / no-go on profit net to Chagee. Used for the appendix pin-to-profit page (earlier extension). Its example values and outlet names are not used; ours are illustrative.

L-chagee-q2-2026-call Chagee (CHA) Q2 2026 Earnings Call, full transcript, TradingKey, 29 Aug 2026 (local copy: 'Chagee (CHA) Q2 2026 Earnings Call_ Profit Margin Improves as Overseas GMV Doubles.pdf', supplied by the team). COO's remarks, interpreted: 'As of the end of June, our total registered members reached 257 million. The repurchase rate among active members remain above 43%, while members who made 2 or more purchases accounted for more than 78% of total orders.' Group-wide, not Singapore. The page's AI summary is not used; the quote is from the transcript.

L-chagee-q1-2025 Chagee Holdings, 'Chagee Announces First Quarter 2025 Unaudited Financial Results', GlobeNewswire, 30 May 2025 (local copy supplied by the team; the team's link https://investor.chagee.com/node/6736/pdf blocks automated access, so not checked by us). 'Chagee's Mobile Mini Program had a total of 192.4 million registered members as of March 31, 2025'; active members = registered members who placed an order at least once in a period. Shows that membership sits on Chagee's ordering app. Group-wide.

P-appstore CHAGEE app, App Store listing (team research, Oct 2026; no link supplied, not checked by us): 'Order and pay digitally', 'Order for self-pickup ahead of time', 'Locate our outlets and their operating hours'. The listing seen was the Saudi storefront; its privacy-policy URL includes 'SGMY' (our reading: Singapore and Malaysia, an inference). Privacy label: Precise Location under 'Data Not Linked to You'; search history not listed.

P-vulcanpost-comeback PRESS - Vulcan Post, Jayme Teo, 2 Oct 2024, https://vulcanpost.com/872176/chagee-singapore-comeback-after-sudden-exit/ : after the reopening, with some customers 'waiting for as long as an hour to receive their orders', 'the brand rolled out in-app ordering and delivery and constantly upgraded the ordering process.' Its 'kiosks' are a store format ('pivoting from their takeaway kiosks for comfortable dine-in, sit-down tea bars'), not ordering kiosks. Quotes checked by us, 22 Oct 2026.

L-sec20f-data Chagee Holdings Form 20-F for FY2025 (same filing as L-sec20f), risk factors: 'Our business generates and processes personal, transactional and other data.' and 'we may not be able to use our membership program to deepen our understanding of our target consumers.' No share of orders by app or mini program, and nothing on search or location data, in the parts we read. Quotes checked by us, 22 Oct 2026.

L-sec20f-member Chagee Holdings Form 20-F for FY2025 (same filing as L-sec20f), Item 3.D risk factor 'We may not be successful in expanding our membership and our ability to take advantage of our membership program may be limited.' Both quoted sentences checked word for word by us against the SEC EDGAR filing, 22 Oct 2026.

C-topic8 OPIM639 course slides, Topic 8 (slides08 Service Platforms), slides 30-36: the economy with network effects, r(x) = 1 - x, f(z) = z, equilibria z' and z". Slide 34 quoted word for word, checked by us against the course file, 22 Oct 2026.

Every Australian input is a scenario assumption

STATUS

Australia worked example, kept separate from every other section: nothing here feeds the main story (slides 01 to 15) or the benefit and nothing in them is changed. Every Australian input is a scenario assumption; no Australian data has been retrieved for this deck. Archetype areas stand in for real suburbs until Census and market data are pulled. Market status: no local Chagee operating history (assumed for this worked example).

CANDIDATE AREAS (ARCHETYPES, SCENARIO ASSUMPTIONS)

A · CBD office core: residents within 1 km 12, workers within 1 km 90, tertiary students within 1 km 15, household income index 115, transit access 0.95, competitors within 500 m 9, rent, a$ a month 22000

B · University precinct: residents within 1 km 18, workers within 1 km 20, tertiary students within 1 km 40, household income index 80, transit access 0.75, competitors within 500 m 6, rent, a$ a month 12000

C · Suburban rail interchange: residents within 1 km 25, workers within 1 km 12, tertiary students within 1 km 5, household income index 95, transit access 0.7, competitors within 500 m 4, rent, a$ a month 10000

D · Outer-suburb shopping centre: residents within 1 km 20, workers within 1 km 5, tertiary students within 1 km 2, household income index 90, transit access 0.35, competitors within 500 m 2, rent, a$ a month 8000

FEATURES AND WHERE REAL VALUES WOULD COME FROM (NONE RETRIEVED)

Residents within 1 km: stands in for walk-in demand from people living nearby; source to pull: national census, usual residents by small area.

Workers within 1 km: stands in for weekday lunch and after-work demand; source to pull: census place-of-work counts.

Tertiary students within 1 km: stands in for the young, app-first customer base; source to pull: university and TAFE enrolment by campus.

Household income index: stands in for willingness to pay for a premium tea; source to pull: census median household income.

Transit access: stands in for footfall past the counter; source to pull: public-transport timetable data, services per hour at the nearest stops.

Competitors within 500 m: stands in for rival bubble-tea and cafe supply (lowers the score); source to pull: business listings or a map-places count.

Rent, A$ a month: stands in for local cost (lowers the score); source to pull: retail lease listings or an agent quote.

Left out: chagee app searches nearby (needs Chagee app users in the market); delivery orders from the area (needs Chagee on the local delivery platforms); distance to the nearest chagee outlet (there is none yet); sales of existing chagee outlets nearby (there are none yet). They need existing Chagee activity. Missing is unknown, not zero: they are left out, never set to zero, so no area is marked down for a gap every area shares.

INPUTS (ASSUMPTIONS.JSON NEW_MARKET)

weights: residents k 1, workers k 1, students k 1, income index 0.5, transit index 1, competitors 0.5, rent month 0.5 relative weight per feature in the suitability score (0-2) (range 0, 2). Our judgement, illustrative. The new-market ranking slide lets you move each; the sensitivity check re-ranks with each weight set to zero and doubled.

sensitivity: low 0, high 2 multipliers applied to one weight at a time. One-at-a-time sensitivity: each weight times zero (dropped) and times two; the rank range is reported per area.

demand scenarios: store.cups per store per year range and value / store.open days per year cups a day: low, base, high. Placeholder scale only: the Singapore per-outlet volume assumption (itself an assumption, not Chagee data) reused for every Australian area. The score does not set cups: without a calibration from real outlets, every area faces the same range.

forecast error: -0.3, 0, 0.3 share by which the pre-opening forecast misses realised cups a day (range -0.5, 0.5). Scenario assumption: before opening, the forecast cannot know actual demand. Under-forecast, exact, over-forecast.

day mix: moderate 104, large 12 busy and event days a year; the rest of store.open_days_per_year are normal days. Scenario assumption for the Australian pilot: busy days = weekend days; event days = a guess at local public holidays and events, not counted. Weights the year-average contribution on the new-market setup slide.

wage per hour: 36 AUD per hour worked, fully loaded (range 30, 45). Scenario assumption, not sourced. Replace with the applicable award rate plus on-costs before use.

machine cost per unit per month: 2000 AUD per station machine unit per month (range 700, 4000). Scenario assumption, not sourced; the Singapore figure is also an assumption and no exchange rate is applied.

price per cup: 8.5 AUD per cup, in store (range 7, 10). Scenario assumption, not sourced. Check menu prices in the pilot city.

utilities per month: 3000 AUD per outlet per month (range 2000, 4500). Scenario assumption, not sourced.

setups: all hand, heading storyboard.split_presets keys compared. Human-only against machine-made drink, as on slide 11; same station speeds and staffing rule (sim/counter.js).

cannibalisation: overlap km 5, max share 0.3 for a second opening: share of a new outlet's cups taken from the pilot, falling linearly to zero at overlap_km. Scenario assumption. Zero for the first opening: no Chagee outlet is in any catchment.

trigger weeks: 4 consecutive weeks a pilot signal must hold before acting. Our judgement, illustrative.

currency: AUD . Every money figure on the new-market slides and the new-market pilot slide is in Australian dollars (A$). No exchange rate is applied to any Singapore figure.

Reused from the Singapore sections, labelled where used: gross margin per cup (storyboard), the per-outlet volume range (store) as the placeholder demand scale, open hours and days, the hourly demand shape, day-type multipliers, station seconds, staffing rule and minimum crews (sim/counter.js, unchanged).

METHOD (SIM/NEWMARKET.JS)

Score: each feature min-max scaled across the areas (turned round for competitors and rent), weighted mean out of 100. **Sensitivity**: each weight dropped, then doubled, one at a time. **Counter**: equipment sized for the base scenario; realised demand reaches the peak hour as a multiplier on the base arrivals; without AI, one crew all day, set for a normal day's planned peak; with AI, the forecast roster for the day type, the forecast missing realised demand by the stated error. **Contribution**: served cups x price x margin, less labour, machine units, rent and utilities, a day; the peak-hour walk-away share is applied to the busiest hours, other hours assumed served. **Worst case**: the lowest year-average contribution over demand scenarios and forecast errors, with AI staffing.

CLAIMS WE DID NOT MAKE

No Australian data and no Chagee data are used. The areas are archetypes, not suburbs. The score is not a sales forecast; the supervised site model is not validated for Australia. No figure on the new-market slides feeds the main story or the benefit.